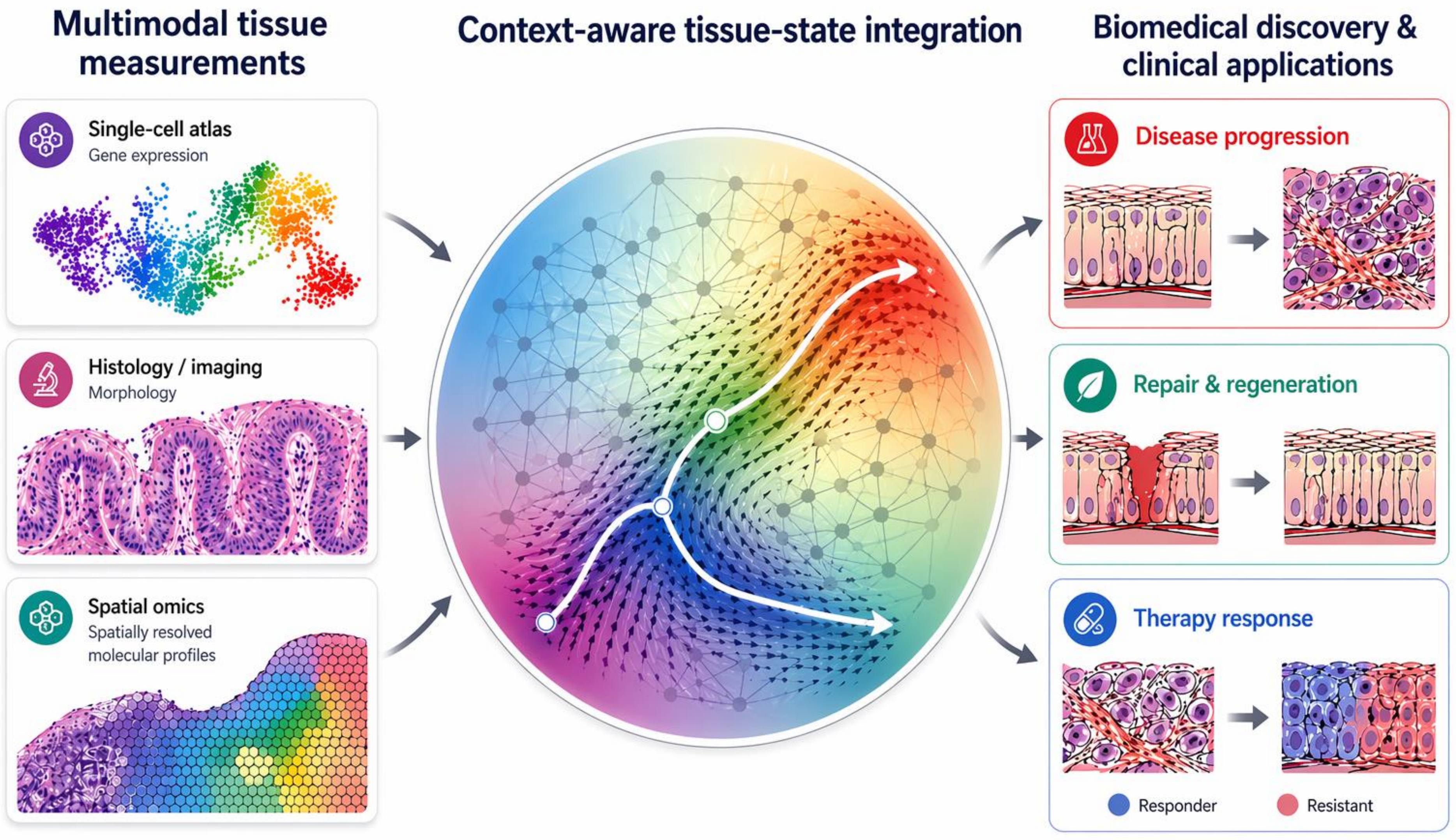
Advanced computational systems biology

Advanced explainable analysis of scRNAseq data

Jovan Tanevski www.tanevskilab.org

Tanevski Lab - The grand challenge - Understanding tissue plasticity

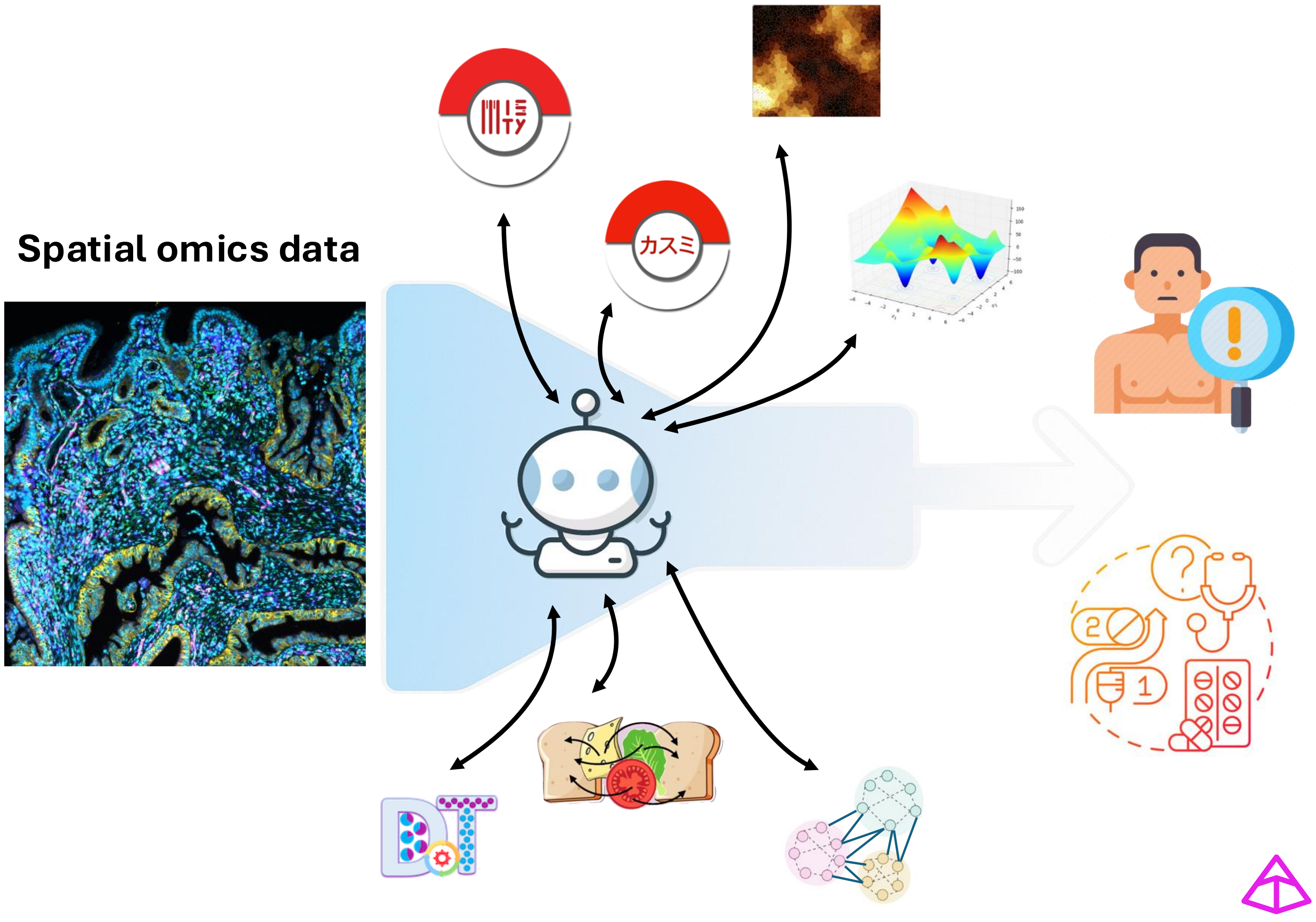


Modeling how tissues transition across health, repair, and disease

Orchestration of specialized models for discovery and clinical translation

Selection and composition

Spatial omics data



Context-aware inference

- Early detection
- Prediction of progression
- Prediction of therapy response
- Biomarker discovery
- Patient stratification

Pydantic AI → langgraph

My part in this course

- 22.06.2026 – Advanced explainable analysis of scRNAseq data
  - Experimental design
  - Batch correction
  - Hypothesis testing
  - Functional enrichment analysis
  - Cell-cell communication analysis

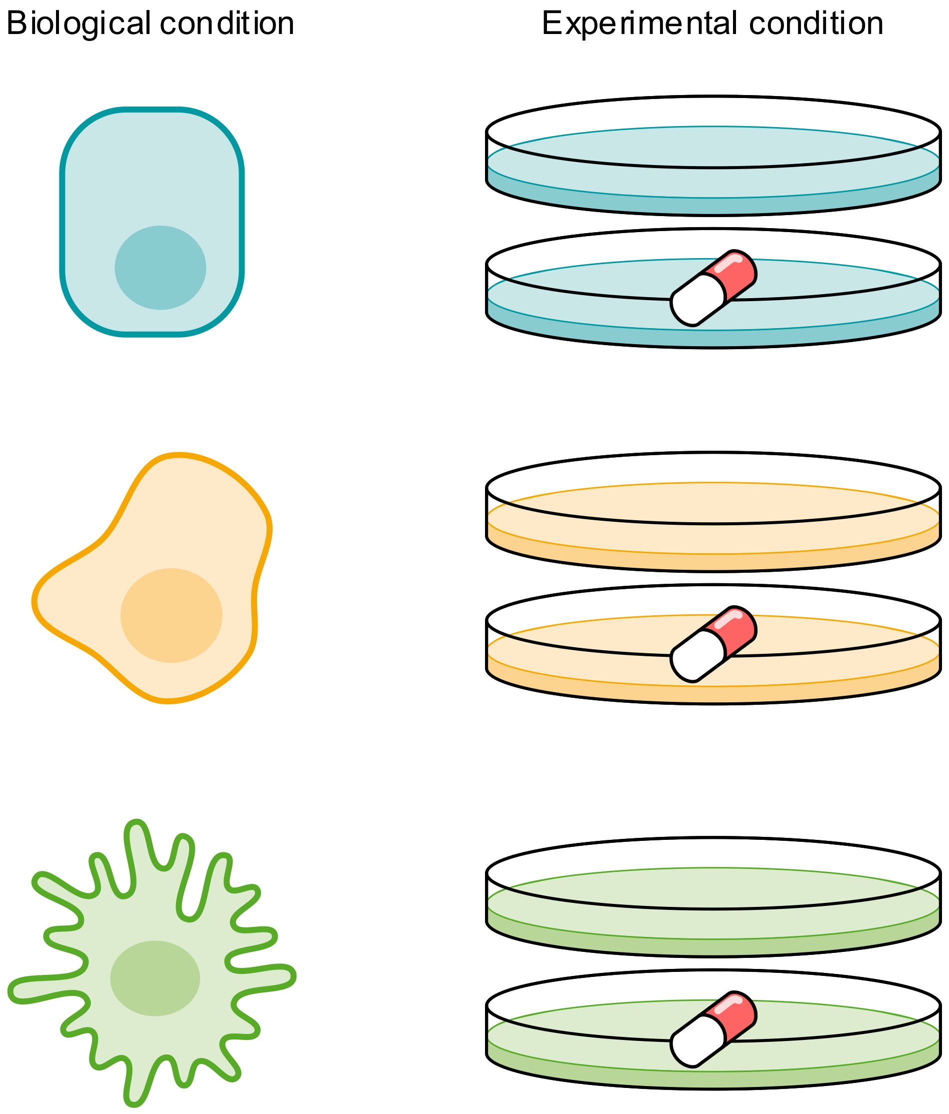
- 25.06.2026 – Advanced explainable analysis of spatial omics data

Acknowledgement

Slides adapted from material used for this course by Julio Saez-Rodriguez and additional material from Nicolas Palacio-Escat and Pau Badia-i-Mompel.

Experimental design and batch effects

Why experimental design is important



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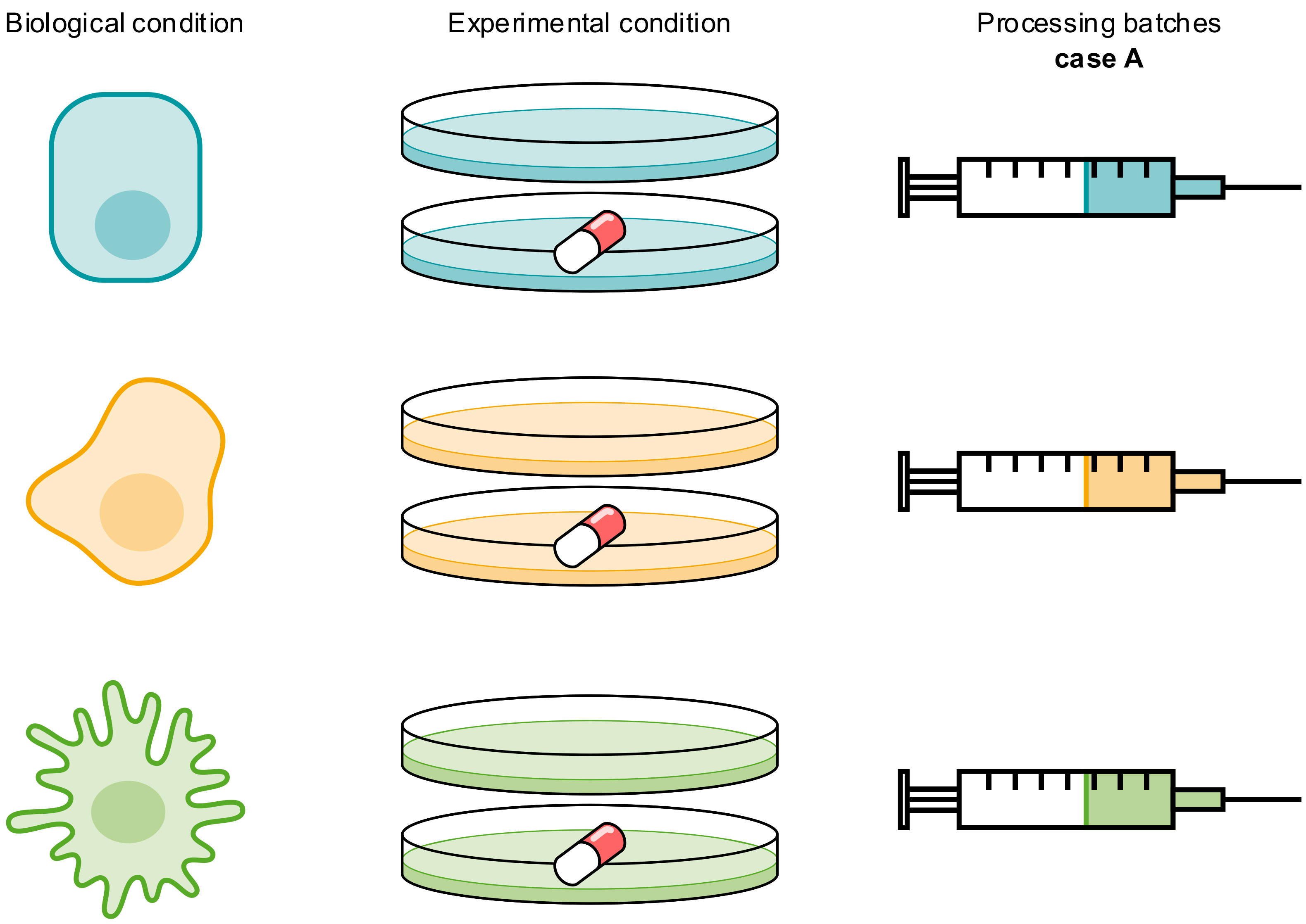

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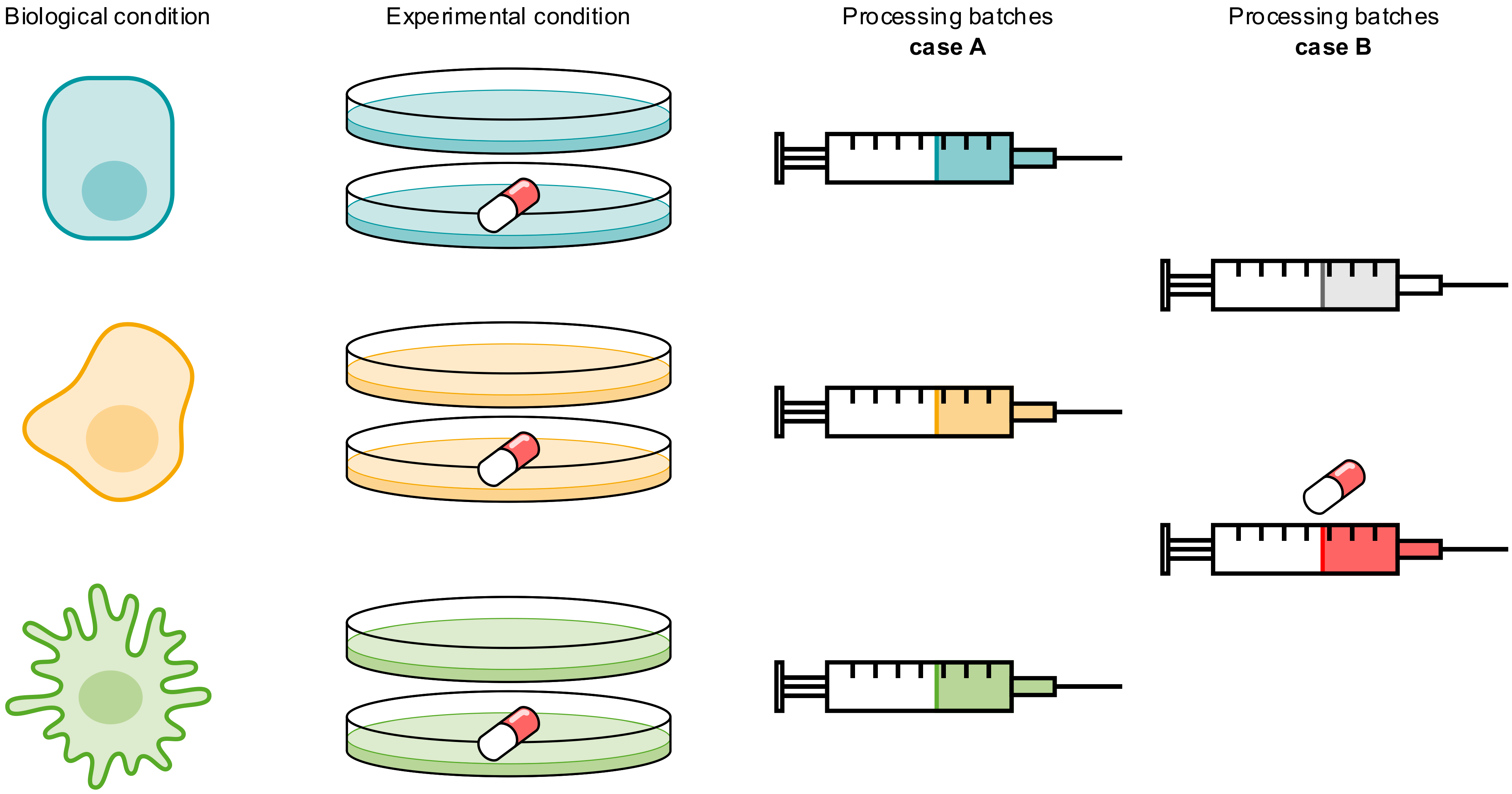

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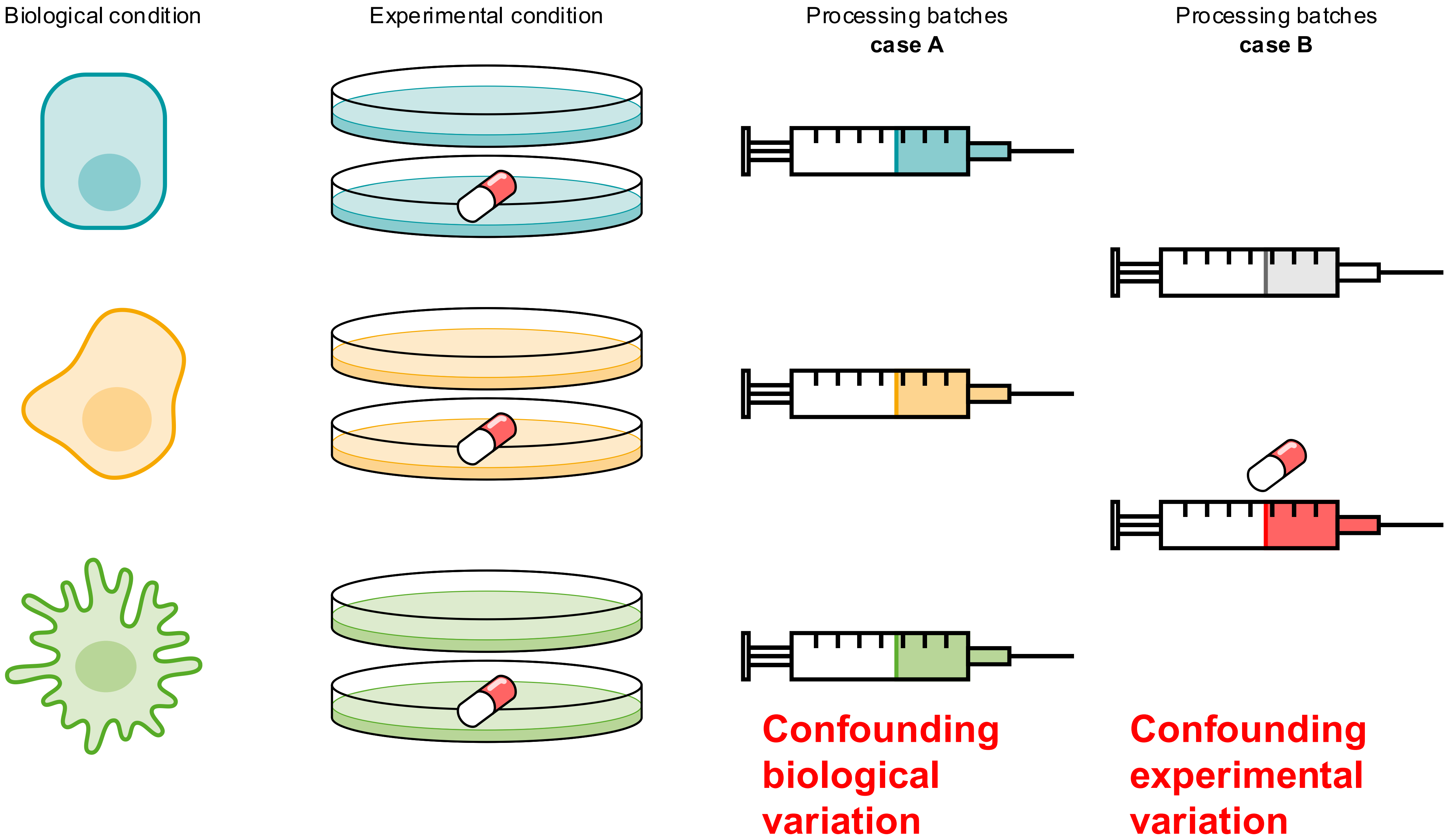
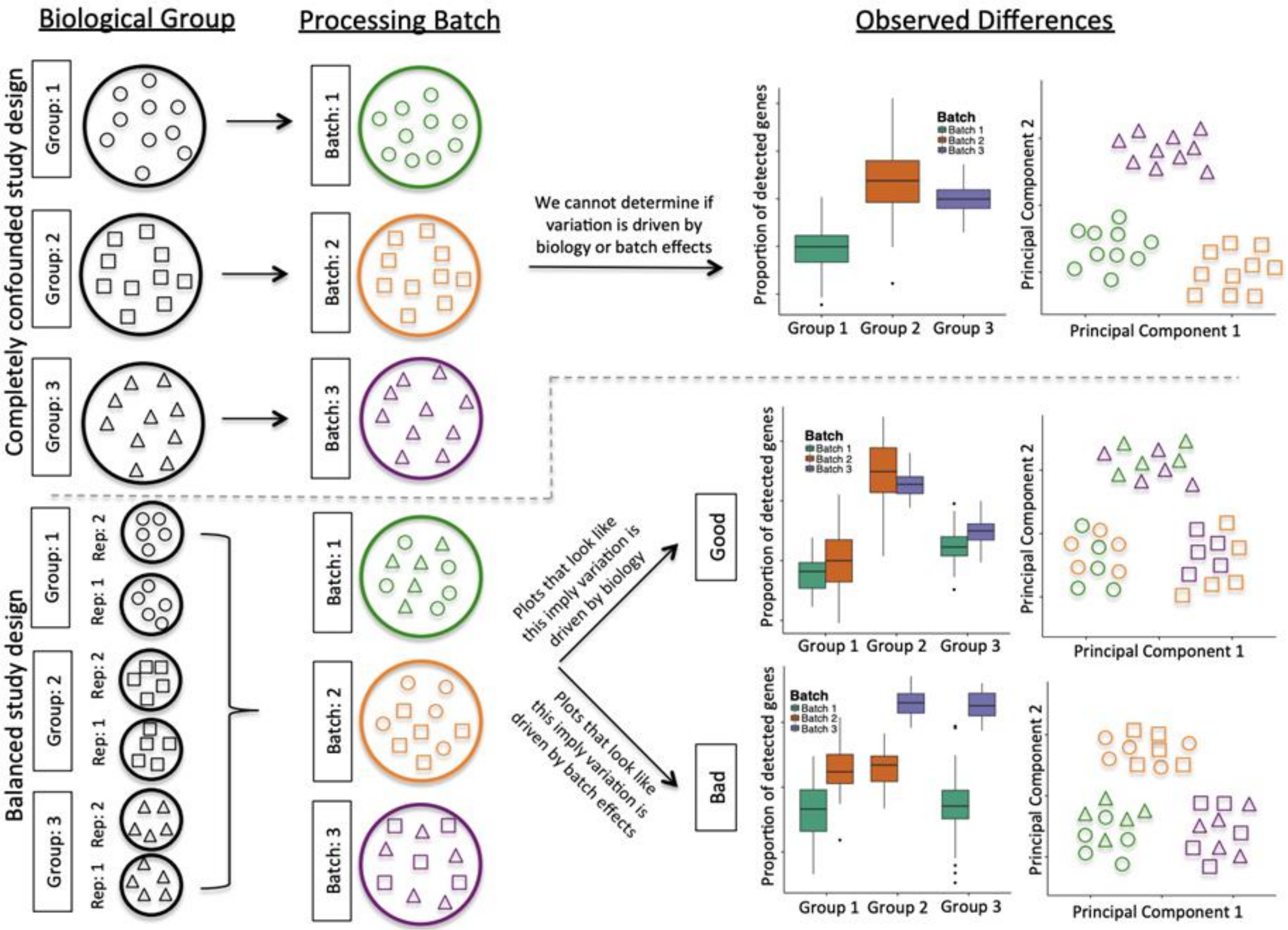
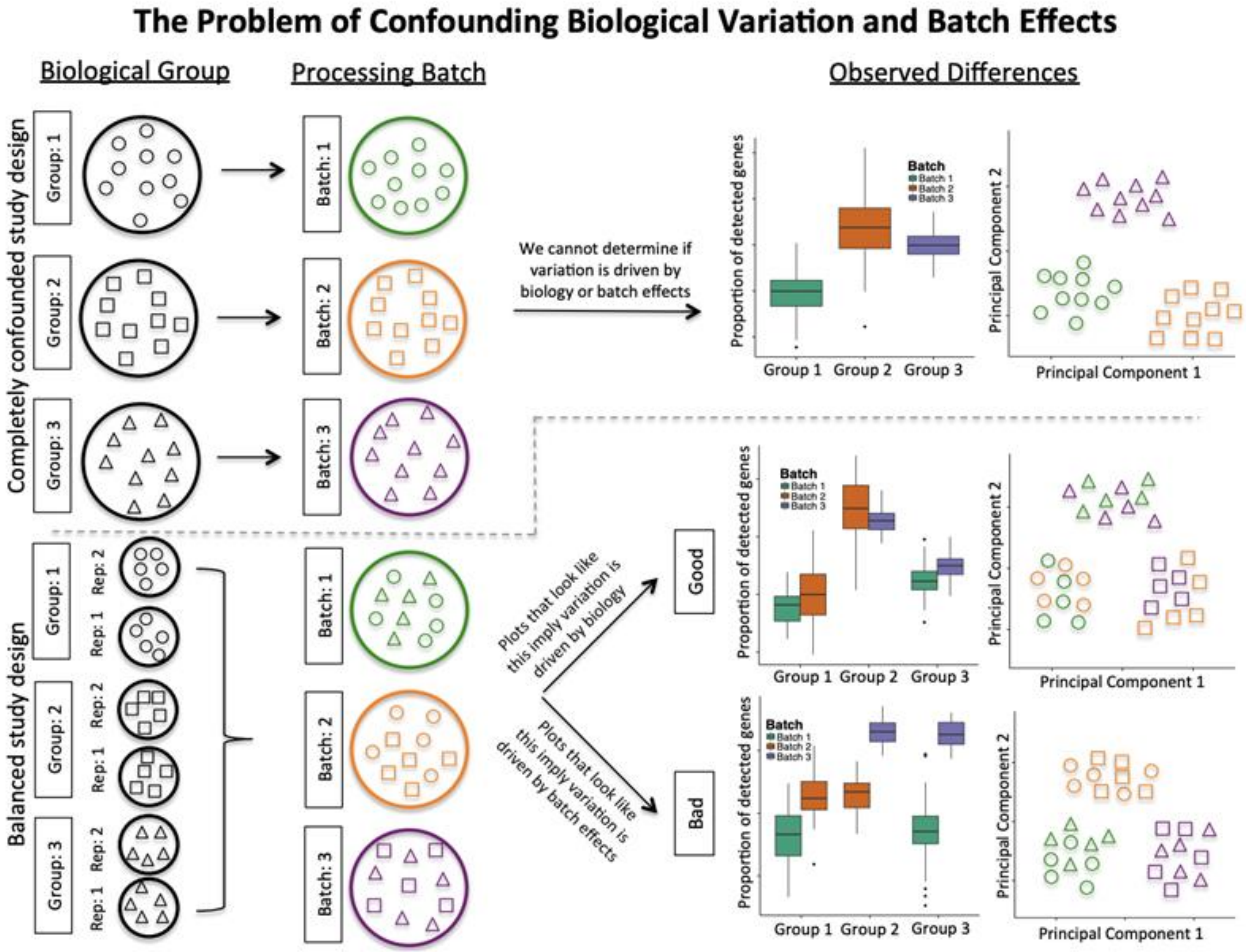


The Problem of Confounding Biological Variation and Batch Effects



https://www.singlecellcourse.org/introduction-to-single-cell-rna-seq.html

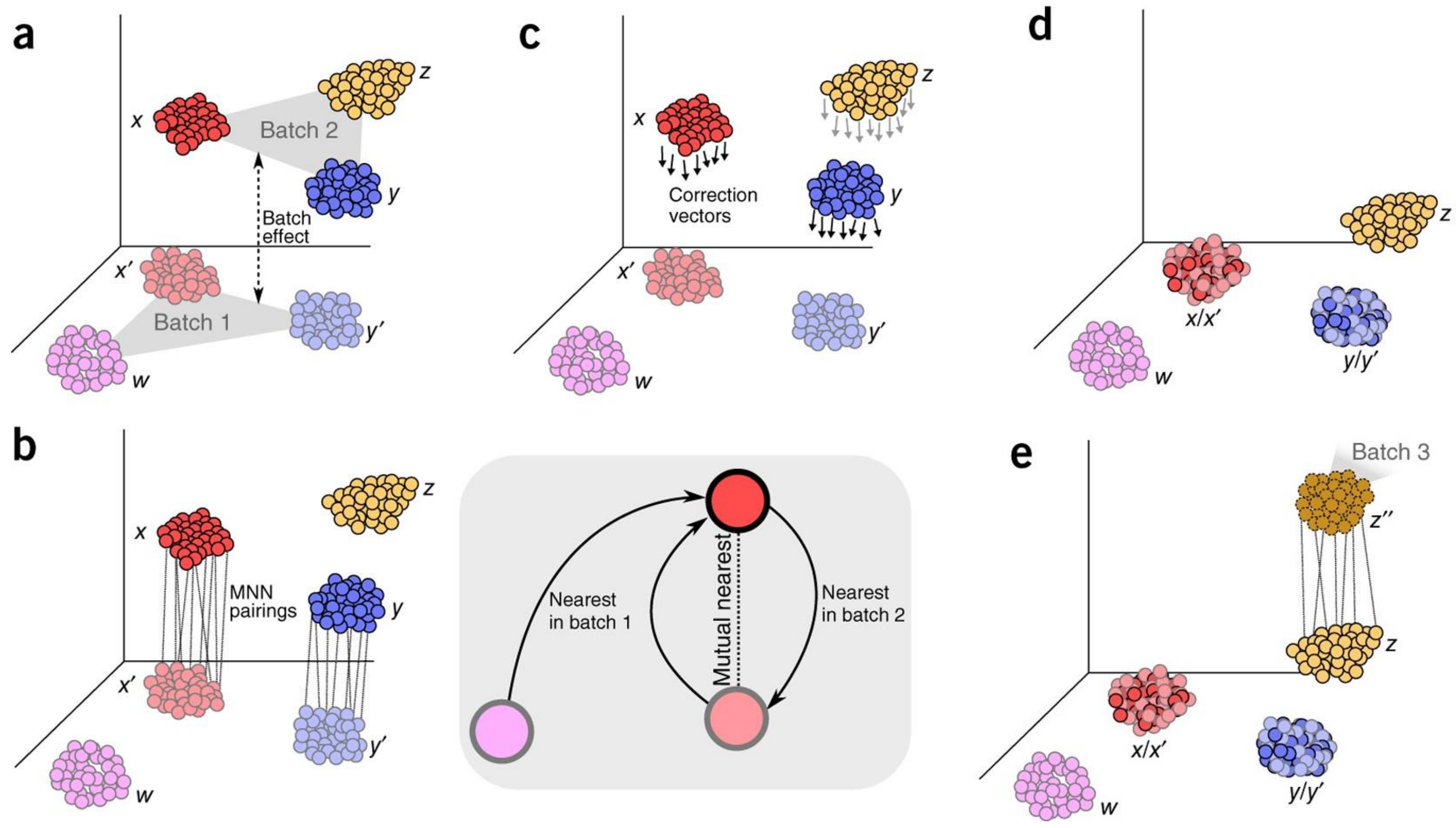
Correcting the batch effect



Many approaches for batch correction
Luecken et al. 2022 Nature Methods

- fastMNN
- Harmony
- scVI
- BBKNN
- Scanorama
- …

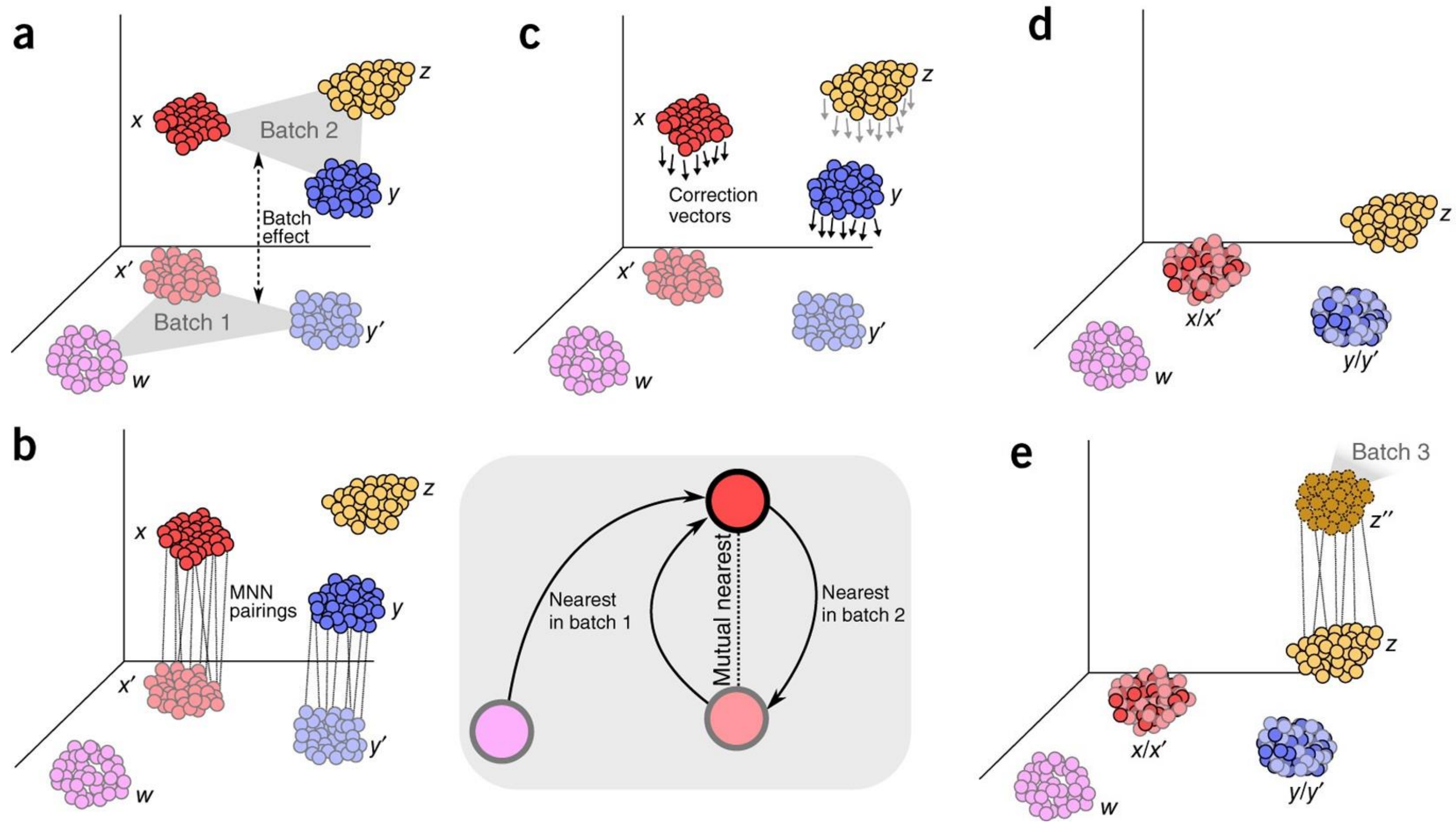
Integration by mutual nearest neighbors



Requirements:
- Batch information
- Common set of genes across batches
- Normalized* and log-transformed* scRNAseq data
- Reduced dimensionality representation (PCA)

Haghverdi et al. 2018 Nature Biotech

Integration by mutual nearest neighbors

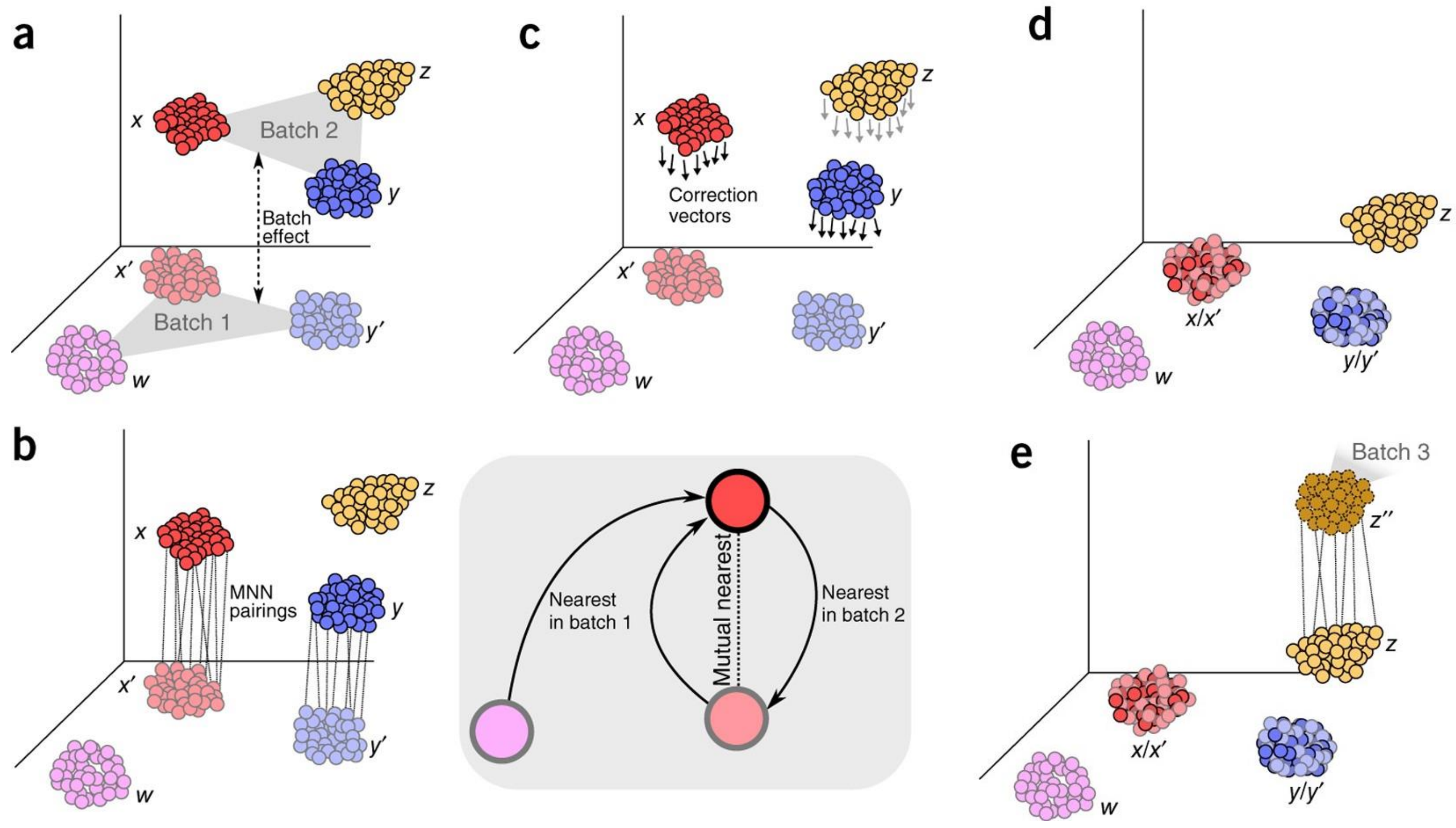


Assumptions:

- Shared biological population (cell types and states)
- Orthogonality - batch effect is orthogonal to and separable from the biological signal
- Relative magnitude of batch effect is smaller than the biological effect

Haghverdi et al. 2018 Nature Biotech

Integration by mutual nearest neighbors



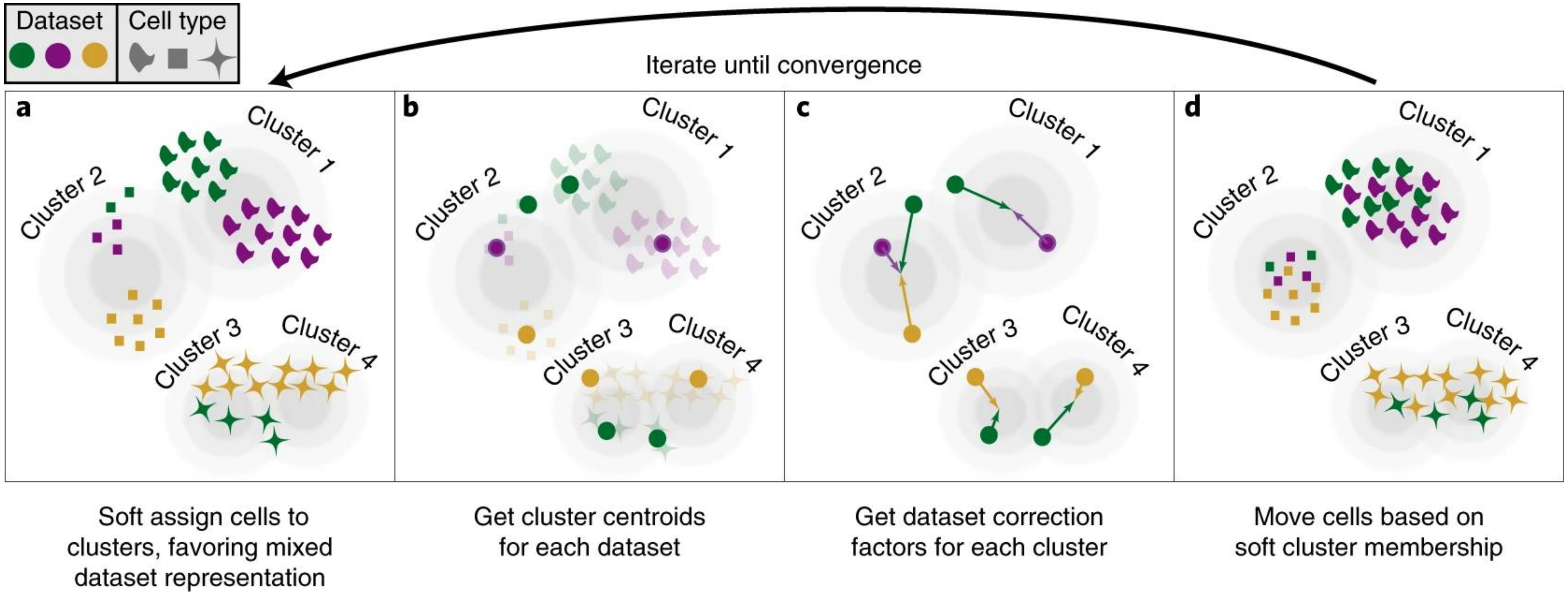
Algorithm:

- Identify mutual nearest neighbors – cell *i* from batch 1 is a MNN of cell *j* from batch 2 if *i* is in the k nearest neighbors of *j* and *j* is in the k nearest neighbors of *i*
- Calculate a vector of differences $v_{ij} = x_i - x_j$ between all pairs (*i*,*j*)
- Average v_{ij} for each cell in batch 2
- Correct each cell of batch 2 by applying the corresponding correction vector

For more than 2 batches, apply this process iteratively

Haghverdi et al. 2018 Nature Biotech

Cell type driven batch correction - Harmony



Korsunsky et al. 2019 Nature Methods

Cell type driven batch correction - Harmony

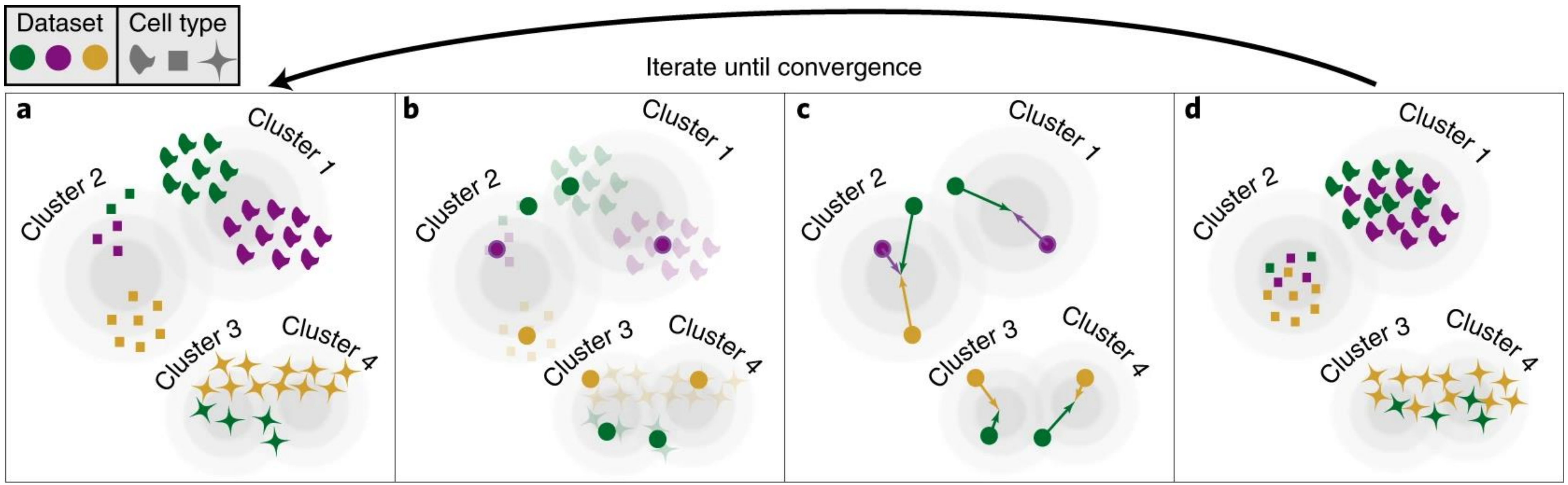
Requirements:
- Batch, dataset, donor assignment information
- Reduced dimensionality representation (PCA) from common set of genes across batches
- Choice of diversity parameter

Assumptions:
- Batch effect is linear in PCA space
- Batch effect is cell-type specific (independent for each cell type)
- Sufficient initial similarity of expression of cell types – cells from the same type across batches in the uncorrected PCA space are closer together than different cell types

Korsunsky et al. 2019 Nature Methods

Cell type driven batch correction - Harmony

Algorithm:



Soft assign cells to clusters, favoring mixed dataset representation

affected by diversity parameter – reward clusters that are geometrically closest but not necessarily homogenous

Get cluster centroids for each dataset

Get dataset correction factors for each cluster

like in MNN but based on the cluster centroids

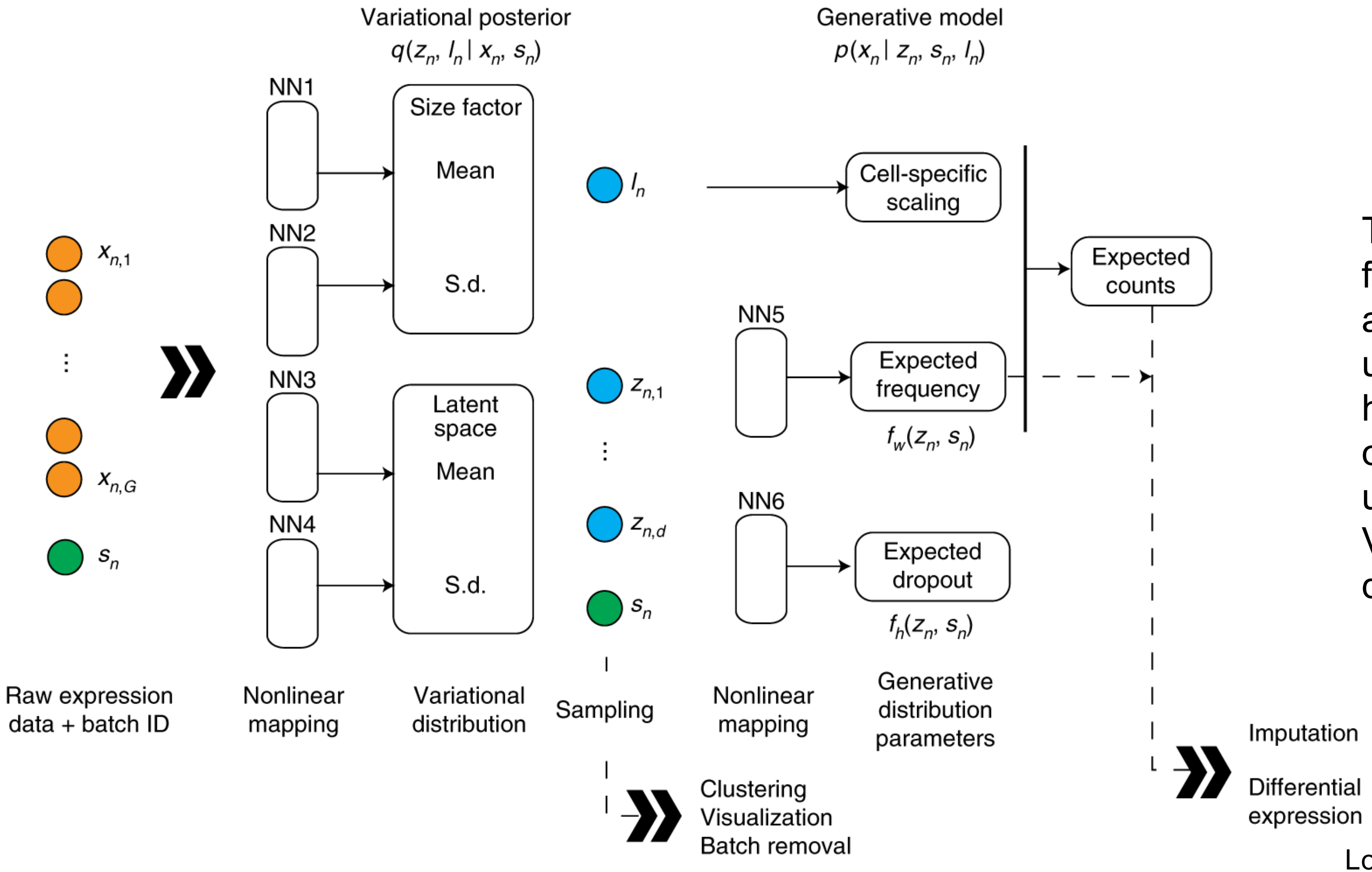
Move cells based on soft cluster membership

probability-weighted (soft clustering) sum of the correction vectors from all clusters

Repeat until convergence

Korsunsky et al. 2019 Nature Methods

Batch correction during integration - scVI



This model represents a fundamental shift from MNN and Harmony: instead of using mathematical heuristics or linear algebra on processed data, scVI uses deep learning (a Variational Autoencoder) directly on raw data

Lopez et al. 2018 Nature Methods

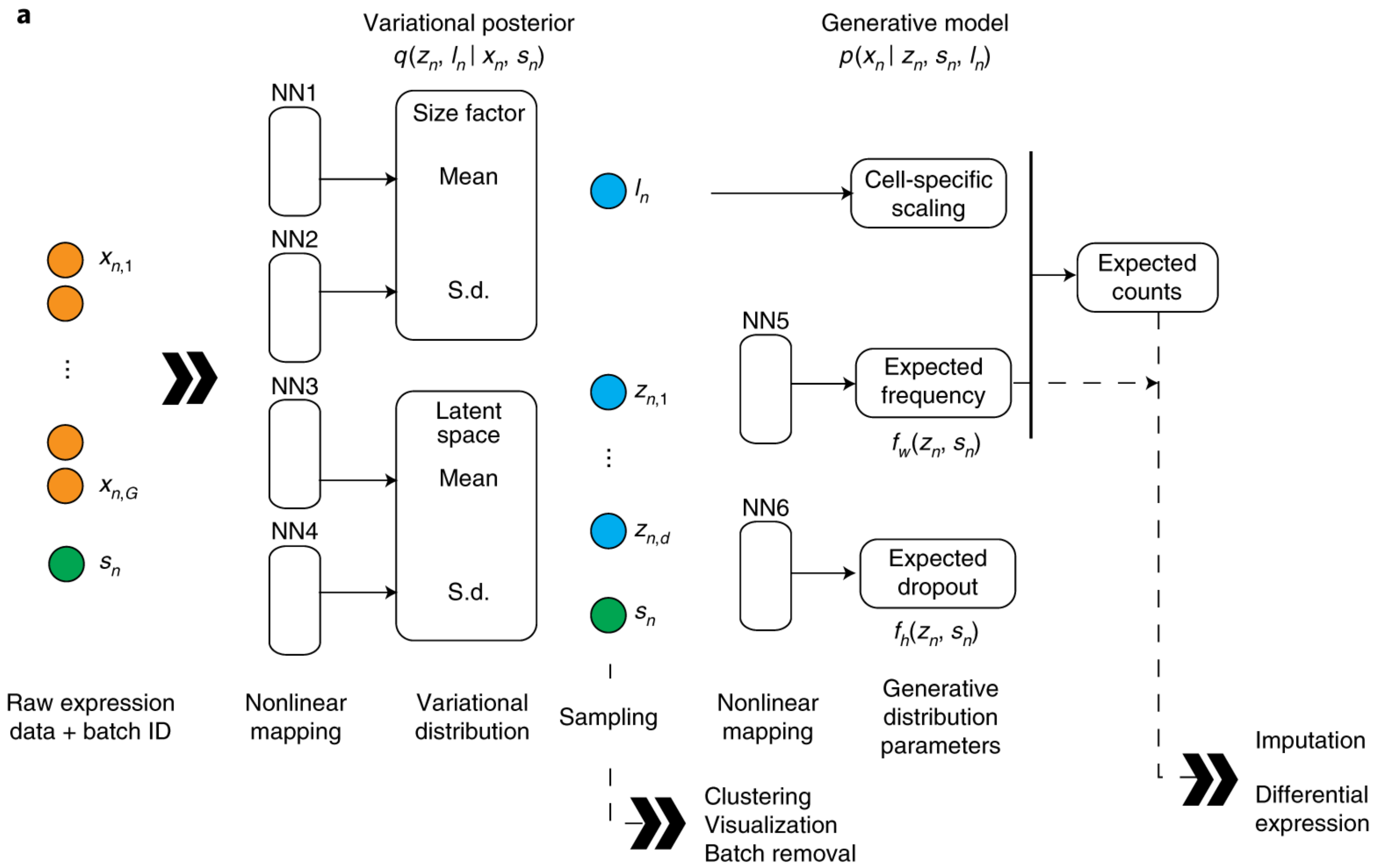
Batch correction during integration - scVI

Requirements:
- Raw, unnormalized count matrices
- Batch assignment for each cell. Can be multivariate.
- Feature selection (1k – 4k HVG)
- Learning hyperparameters – layers, rate, latent space dimension

Assumptions:
- Generative count distribution – NB or zero-inflated NB
- Non-linear biological manifold. Complex models are needed to capture complex gene-gene interactions
- True biological state can be decoupled from technical effects – even non-linear ones

Lopez et al. 2018 Nature Methods

Batch correction during integration - scVI

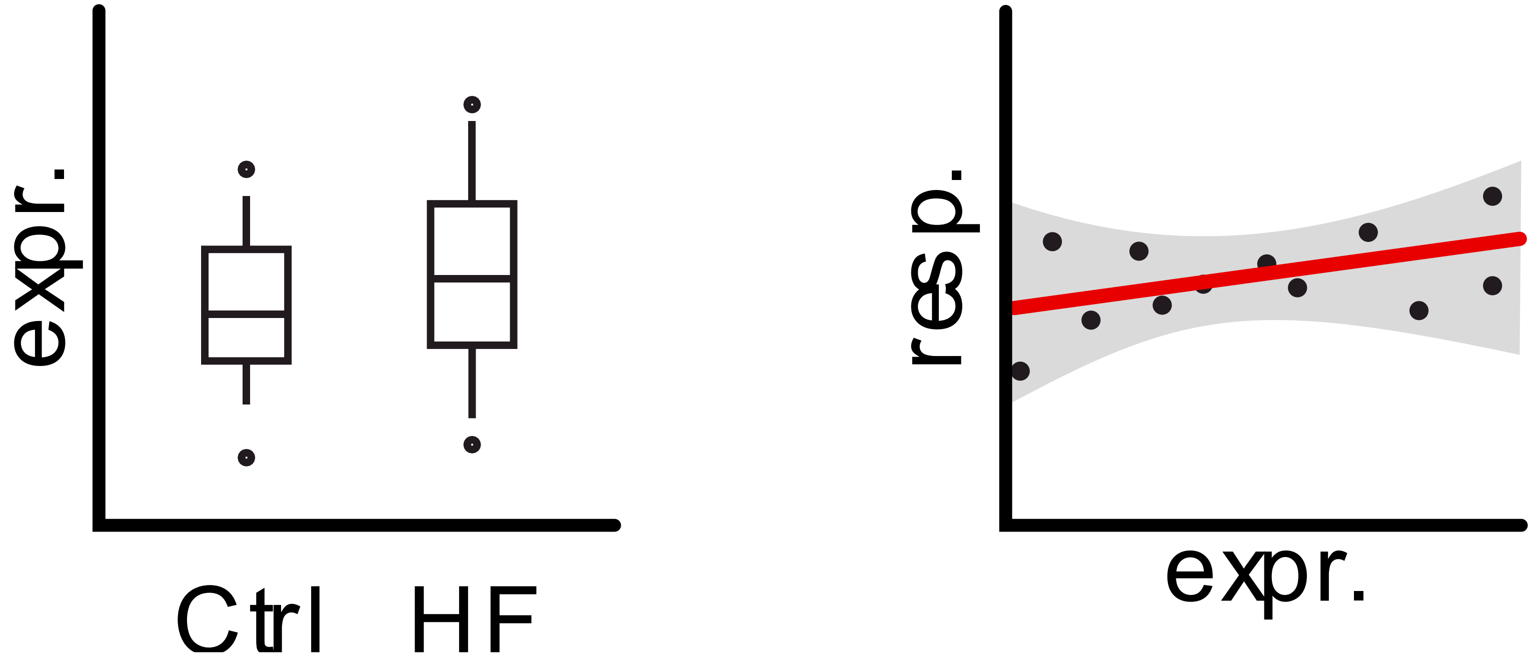


Algorithm:

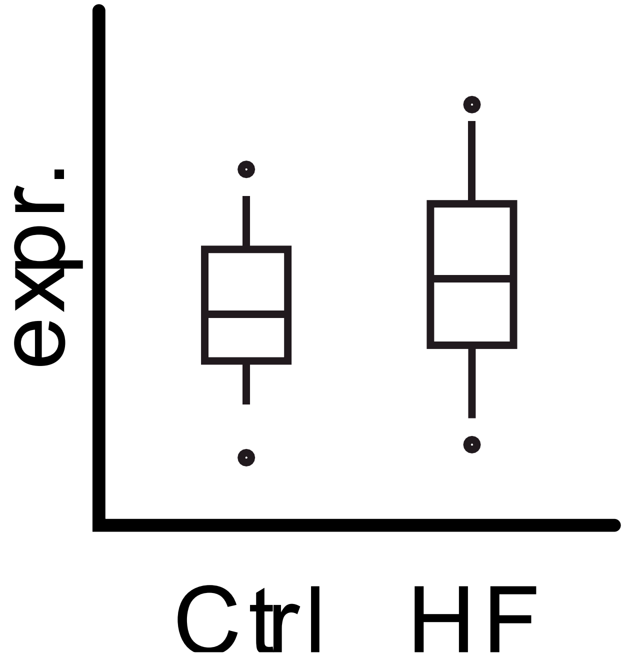
- Encoder – feed raw counts into a NN and batch (meta) data – output mean and variance of distributions of size factor and the latent space
- Sampling the latent space – randomly select a point from the learned latent space to generate "clean" cell identity *(used ultimately for UMAPs, clustering, trajectories, etc.)*
- Decoder – Recreate the original cell expression – concatenates the batch data – answers: if I have this "clean" identity, what would the cell look like under this particular batch effect
- Training loss function – quality (KL) of reconstruction (regularized)
- Iterate in small batches with stochastic gradient descent until convergence
- Discard the decoder part and use representation from step 2

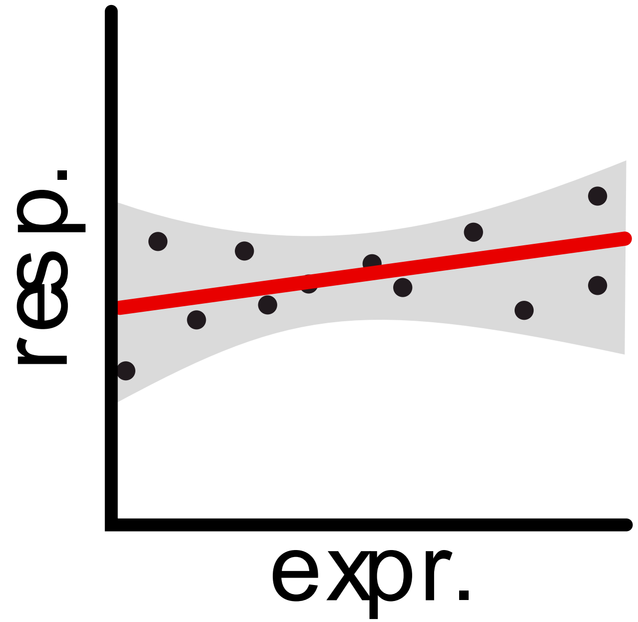
Hypothesis testing

Statistical inference – from samples to population



We observe:
- Expression of a gene differs between Ctrl and HF
- Response to treatment correlates to expression of a gene





We observe:
- Expression of a gene differs between Ctrl and HF
- Response to treatment correlates to expression of a gene

But:
- Would we observe this in another cohort?
- Does this hold for the entire (unknown) population?

Are these observations significant?

Hypothesis testing

- Question:
  - Is there a difference between Ctrl and HF in the expression of gene A?
  - Is there a correlation between treatment effect and expression of gene B?

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- Hypothesis:
  - Null hypothesis (H_0)
    - Expression of gene A is no different between Ctrl and HF
    - Correlation between effect and expression is 0
  - Alternative hypothesis (H_1)
    - Expression of gene A differs between Ctrl and HF
    - Correlation between effect and expression is not 0

Hypothesis testing

- Question:
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  - Alternative hypothesis (H_1)
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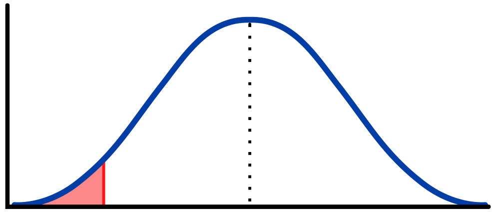
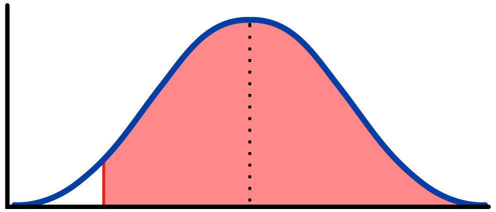
- Test statistic: A value that can be computed from the data (depends on variable types and distribution) and we can compare to a known distribution

What is a p-value?

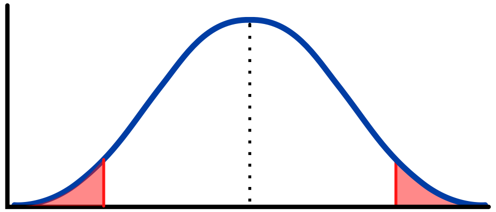
- Probability of obtaining…:



- Smaller (one-sided, lower tail)



- Larger (one-sided, upper tail)



- More extreme (two-sided/tailed)

… value of the test statistic if H_0 is “valid”

Interpreting p-values

- Define a significance level (α)
  - $p < \alpha$:
    - H_0 can be rejected
    - H_1 Statistically significant
  - $p > \alpha$:
    - H_0 cannot be rejected
    - H_1 can be due to statistical fluctuations

- $\alpha = 0.05$ has become standard, but is totally arbitrary

The t-test

- Test for comparing means (averages)
  - **One-sample:** Comparing the mean of a sample against the population
  - **Two-sample:** Comparing the mean of two samples
    - Unpaired (independent): e.g. gene A vs. gene B
    - Paired (dependent): e.g. gene A before and after treatment (aka repeated measures)

When can we apply a t-test

- Conditions to be fulfilled to apply:
  - **Normality**:
    - The data must be normally distributed (can be tested by Q-Q plot or with Shapiro-Wilk test)
    - Note: This applies to the **sample we are testing**, not the *experimental* sample
    - When not normal, apply other non-parametric test
  - **Variance** of the samples must be *equal,* otherwise Welch's t-test
  - **Independence** - values in a sample should not be influenced by the other

Functional analysis

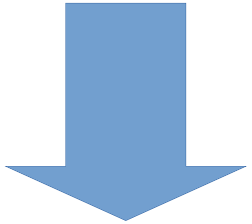
Standard set up to study molecular mechanisms


Condition A   Condition B

Standard set up to study molecular mechanisms

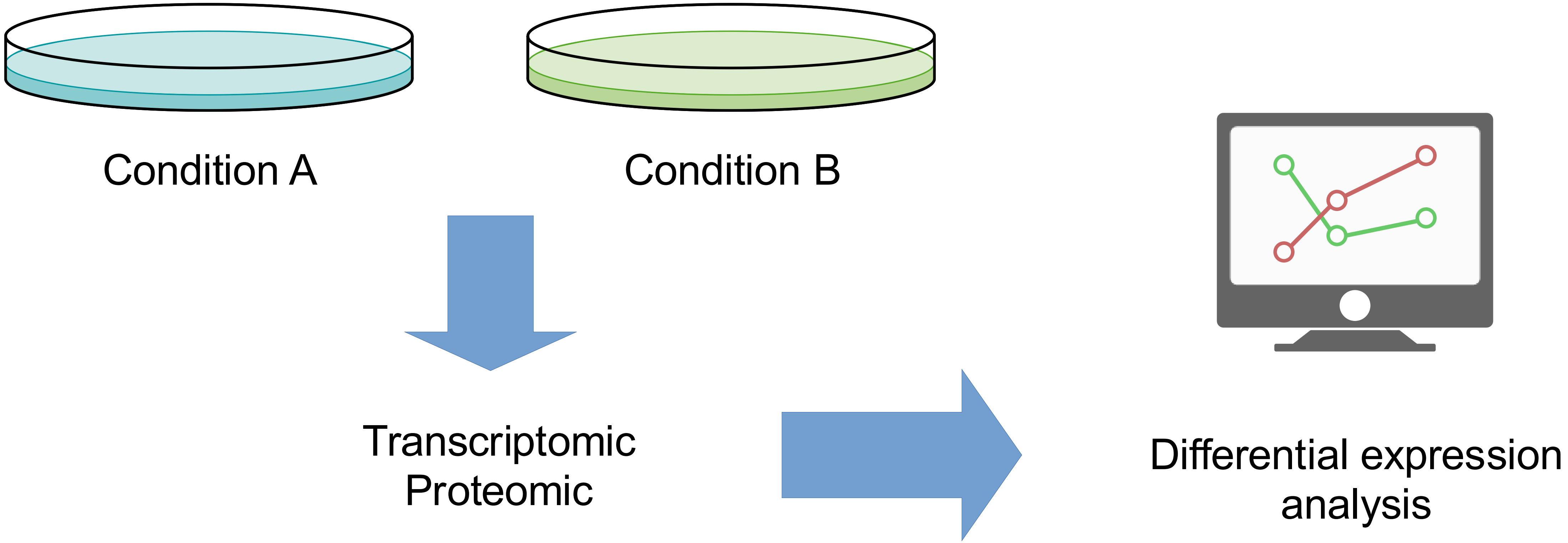
Condition A

Condition B

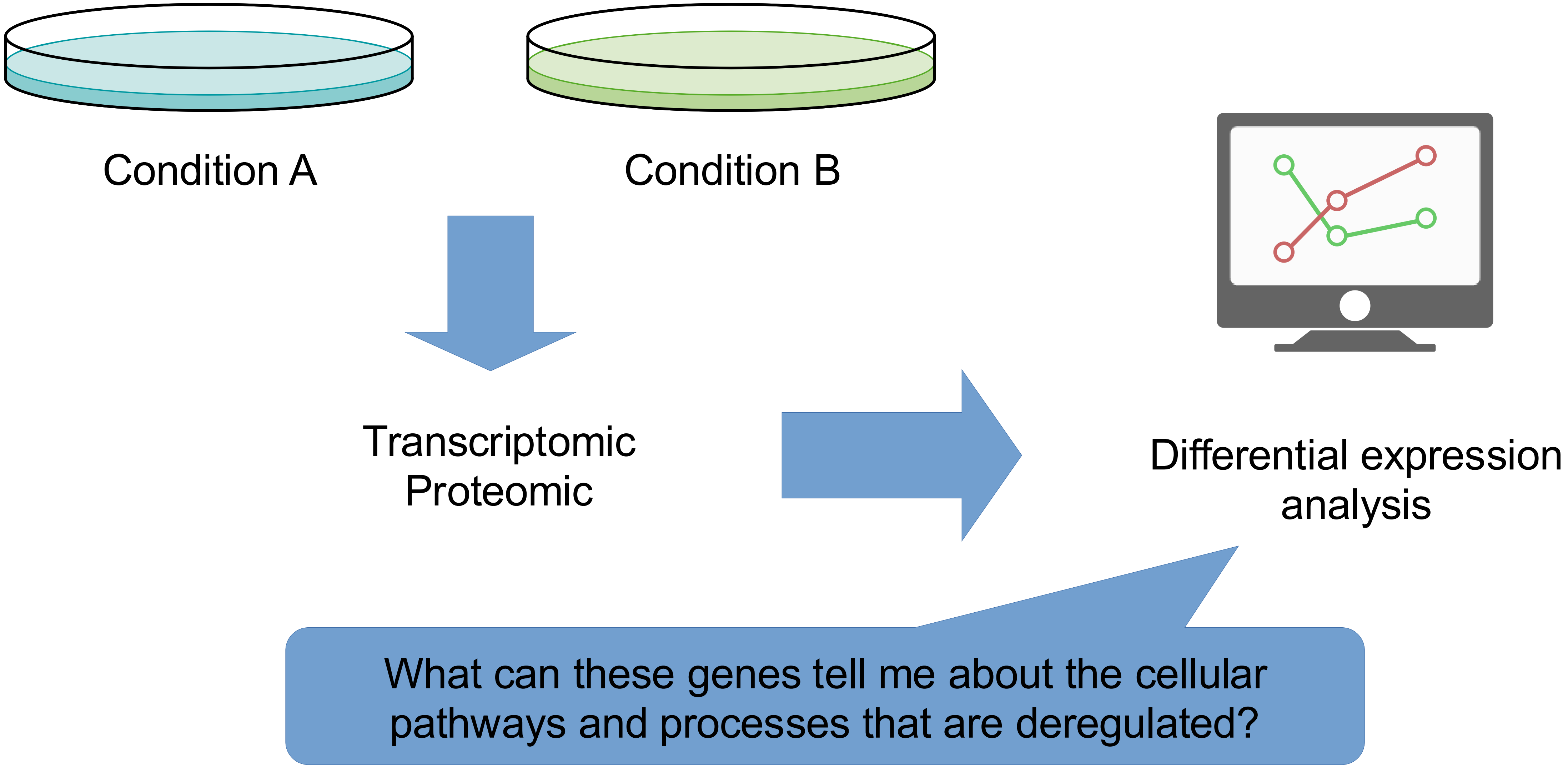


Transcriptomic
Proteomic

Standard set up to study molecular mechanisms

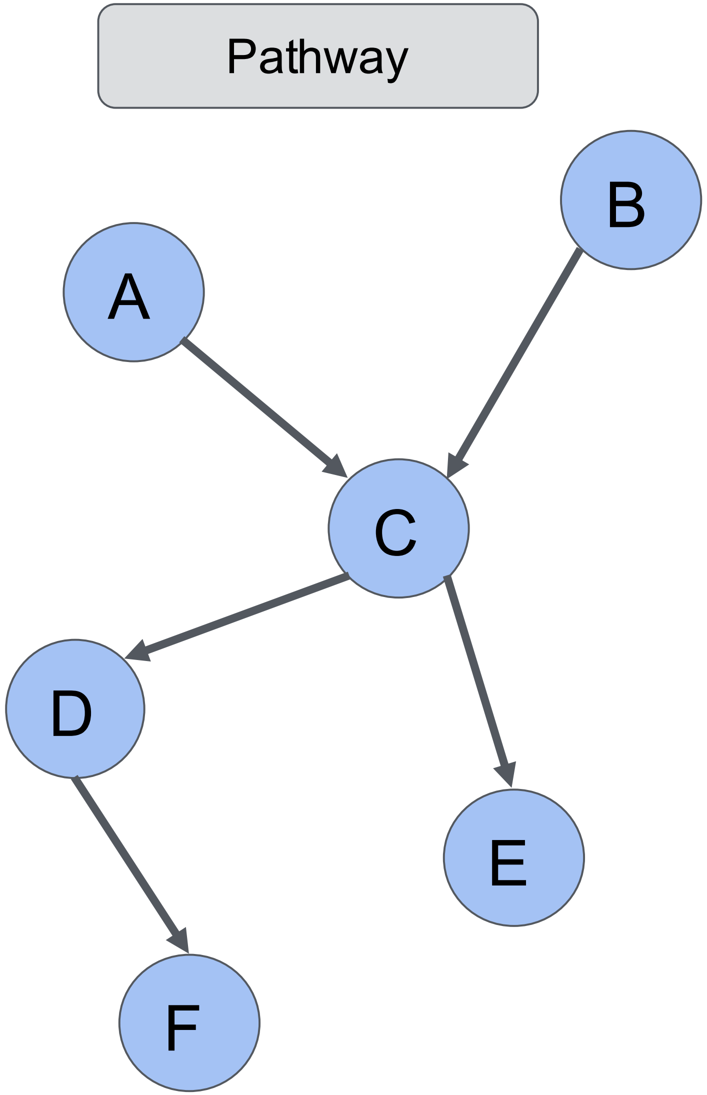


Standard set up to study molecular mechanisms



Pathways: from network to set representation

Classic network view of pathway



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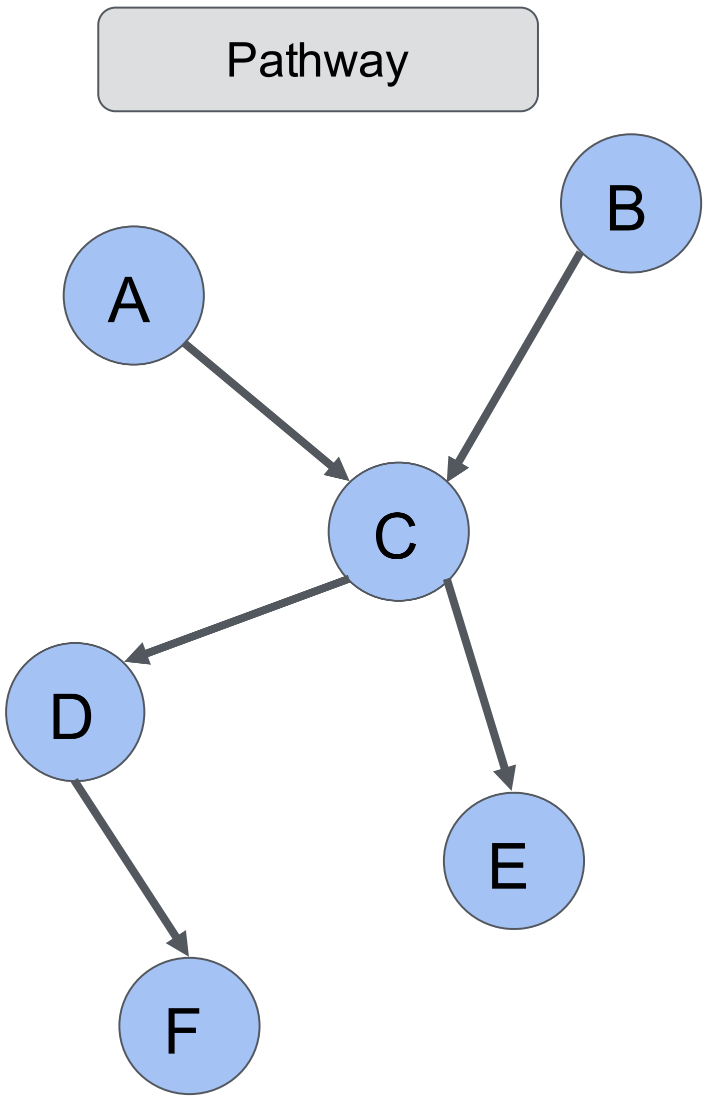
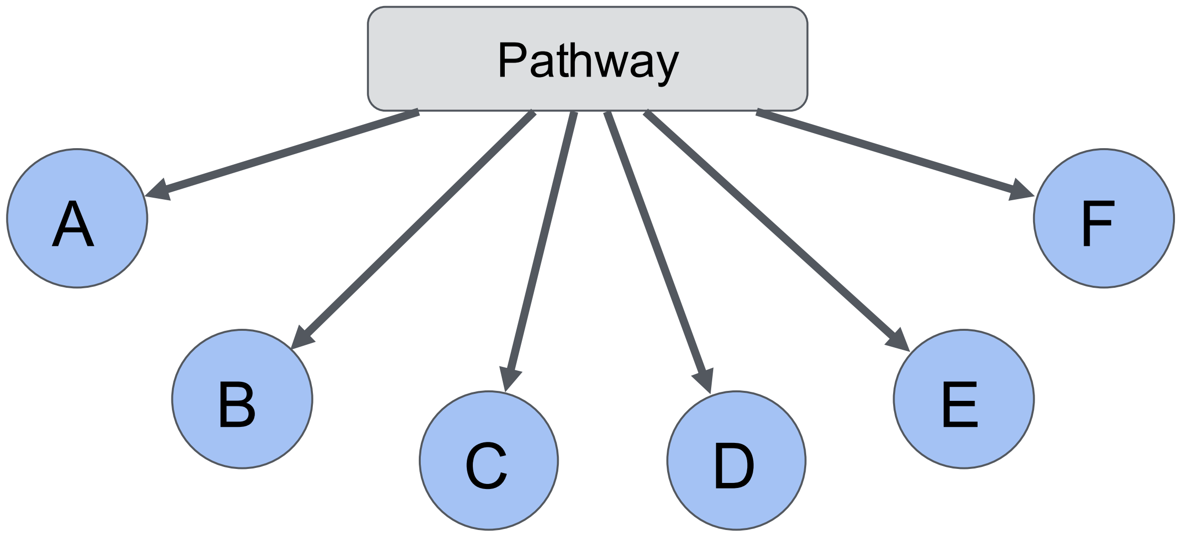


Representation of pathway as set



Over-Representation Analysis (ORA)

Considering the differentially expressed genes between two conditions, are there groups of genes (gene sets) that contain significantly more differentially expressed genes than expected?

Step 1 : what is the proportion of deregulated/non-deregulated genes in pathways for the experiment.

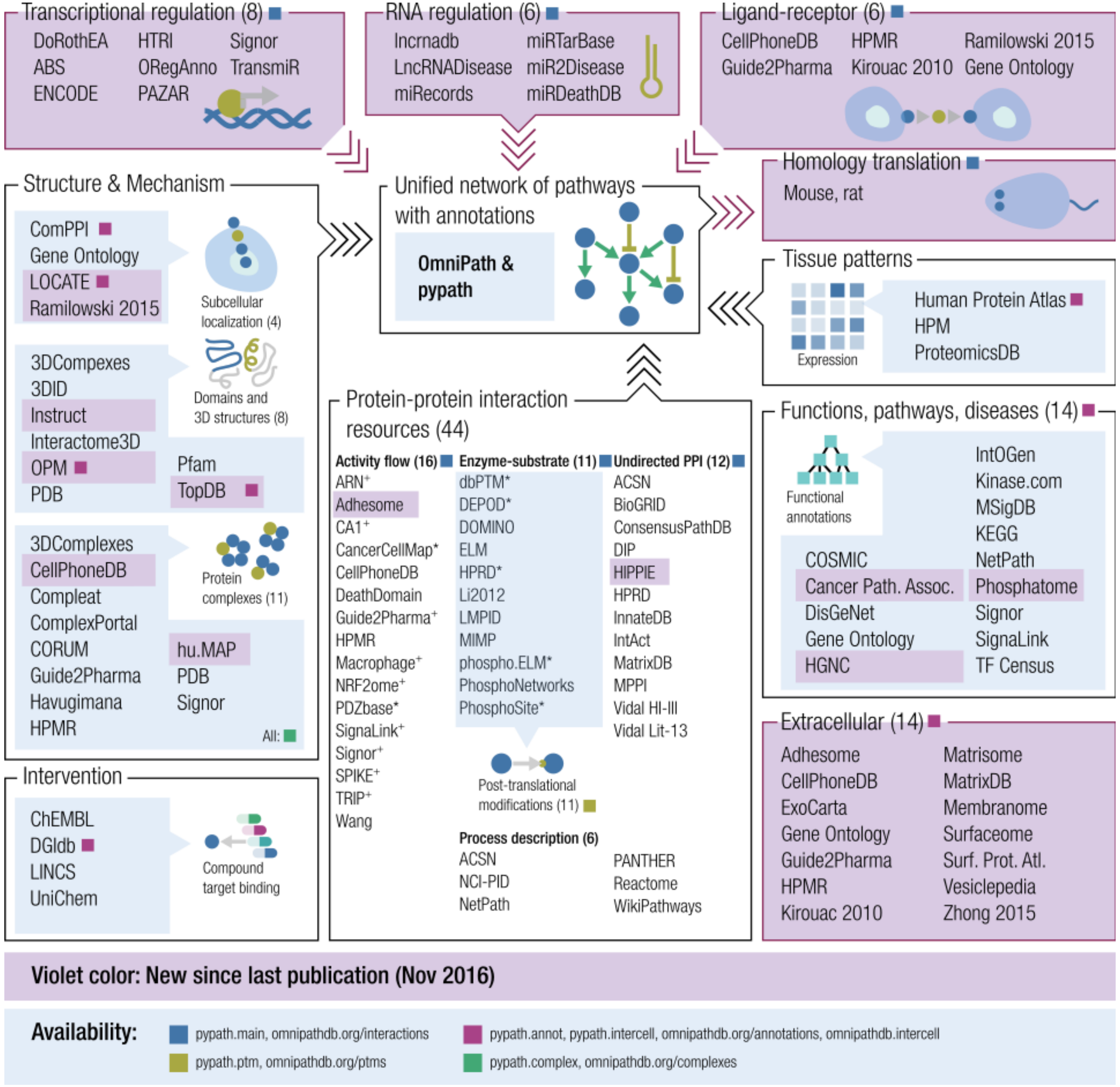
Over-Representation Analysis (ORA)

Considering the differentially expressed genes between two conditions, are there groups of genes (gene sets) that contain significantly more differentially expressed genes than expected?

Step 1 : what is the proportion of deregulated/non-deregulated genes in pathways for the experiment.

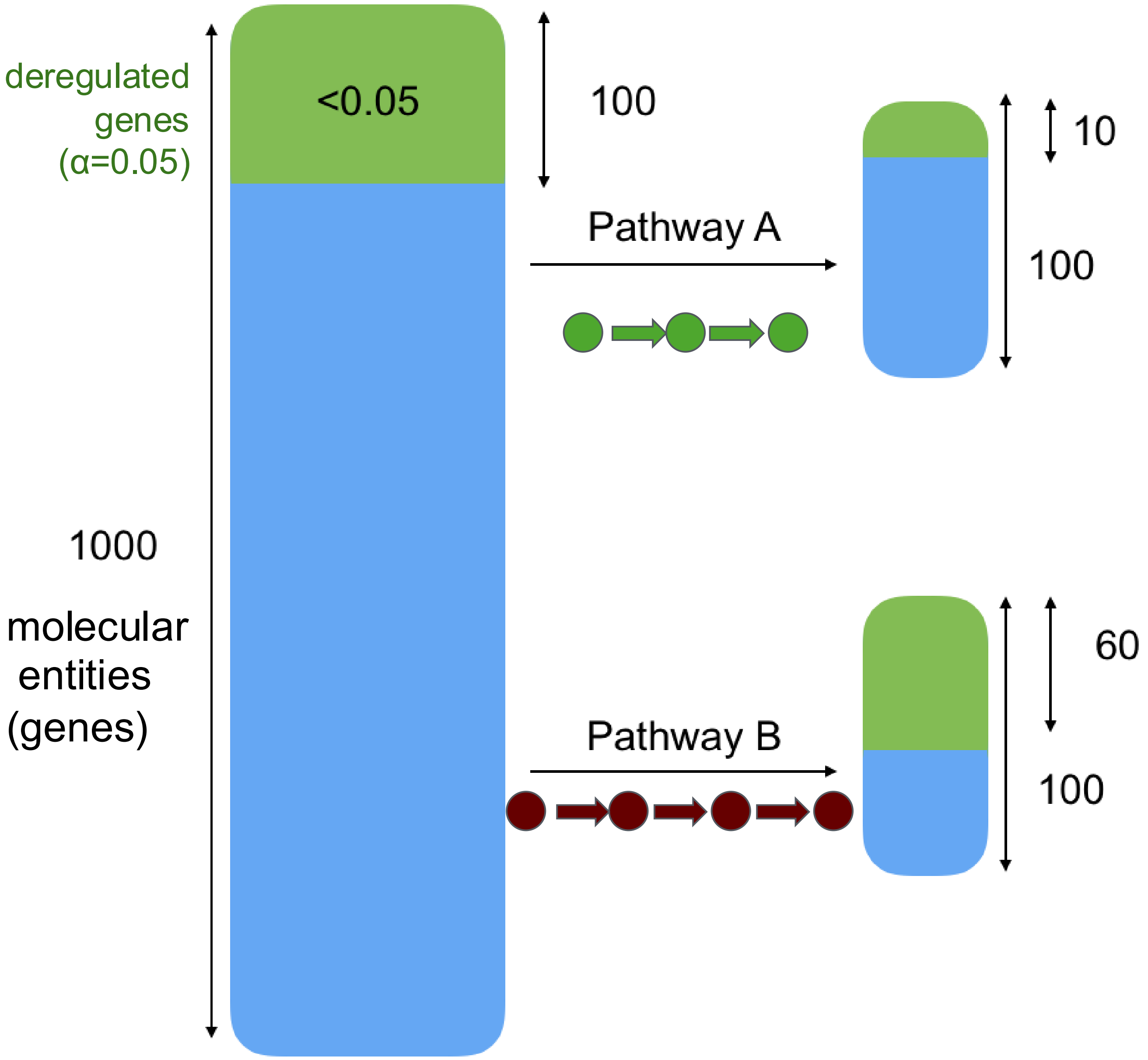
Step 2 : are the proportions in pathway SIGNIFICANTLY different from what is expected.

But first, where to get prior knowledge



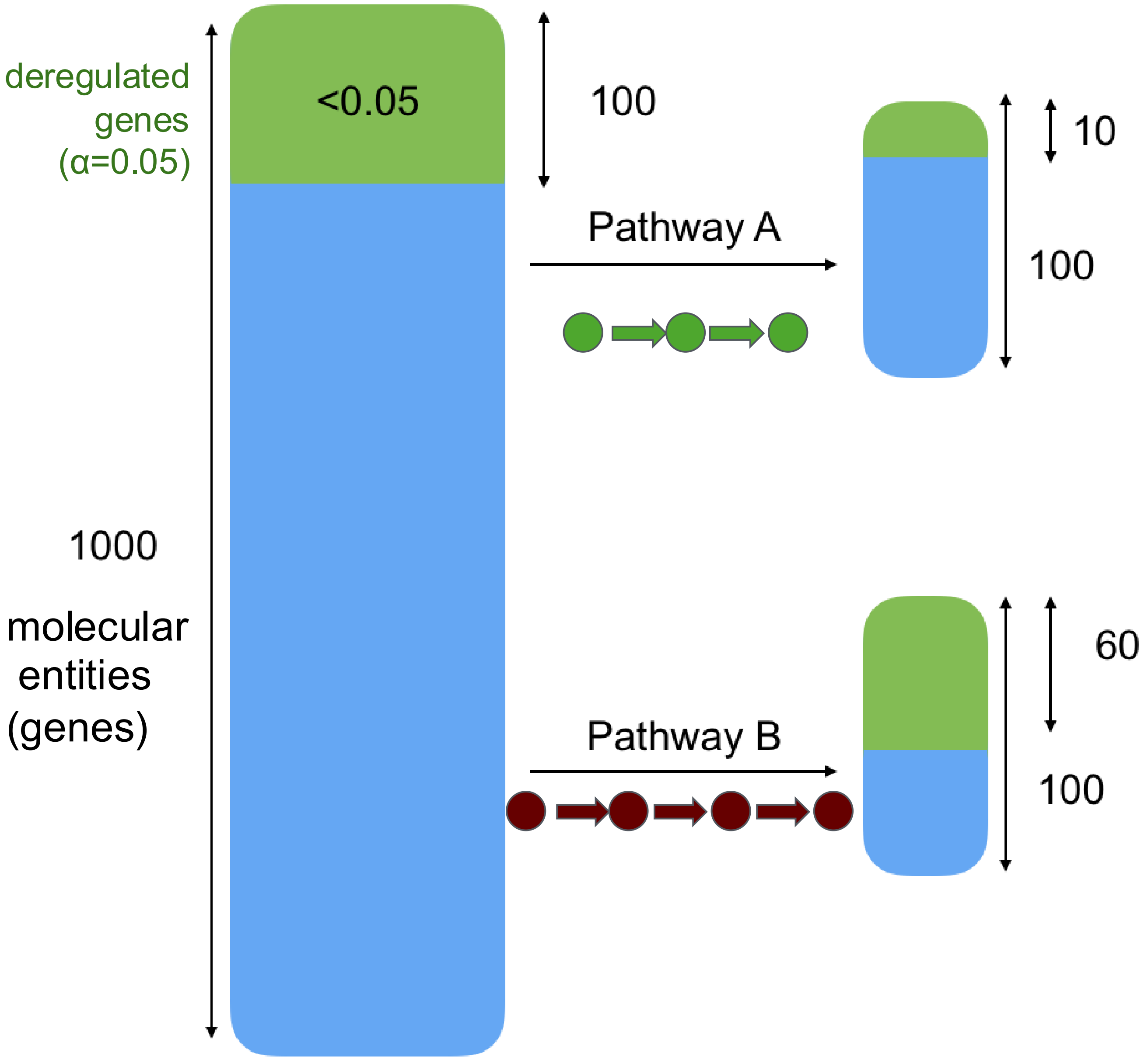
Turei et al. 2016 Nature Methods
Turei et al. 2021 Mol Syst Biol
www.omipathdb.org

Over-Representation Analysis (ORA)



expected proportion of deregulated molecular entities in a pathway:
100/1000 = 0.1

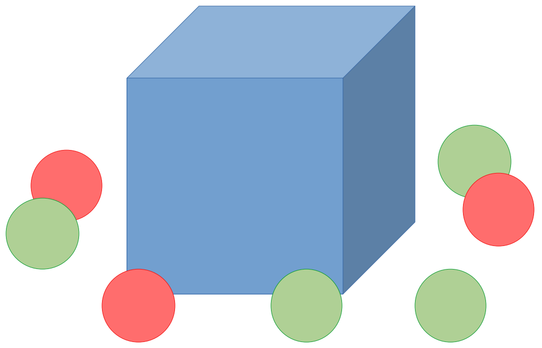
Over-Representation Analysis (ORA)



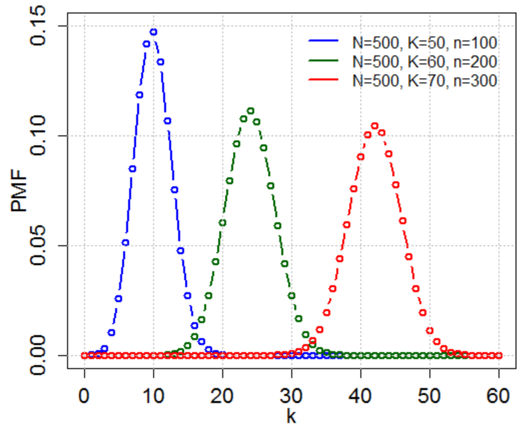
expected proportion of deregulated molecular entities in a pathway: $100/1000 = 0.1$

How likely is it to be so far from the expected proportion by chance?

Sidetrack: Hypergeometric distribution

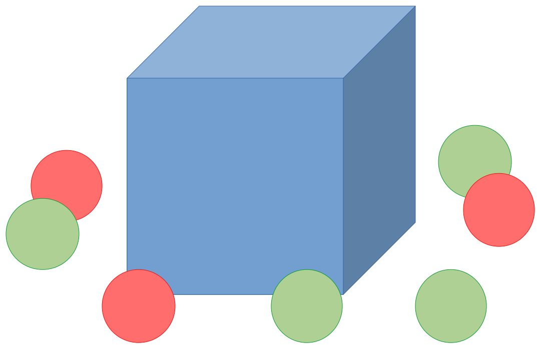


- Discrete probability distribution:
  - Describes the chances of *k* successes in *n* draws
  - For a population of *N* elements of which we have *K* with a given feature



https://en.wikipedia.org/wiki/Hypergeometric_distribution#/media/File:HypergeometricPDF.png

Sidetrack: Hypergeometric distribution



- Discrete probability distribution:
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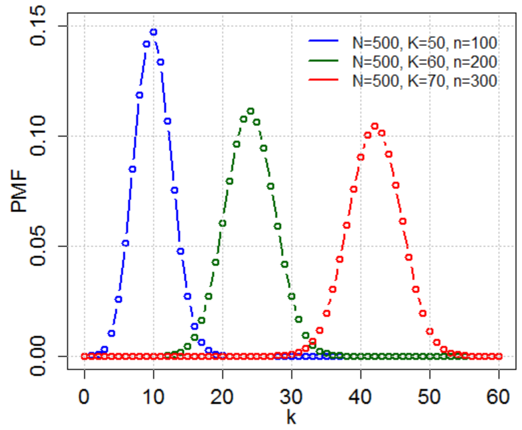
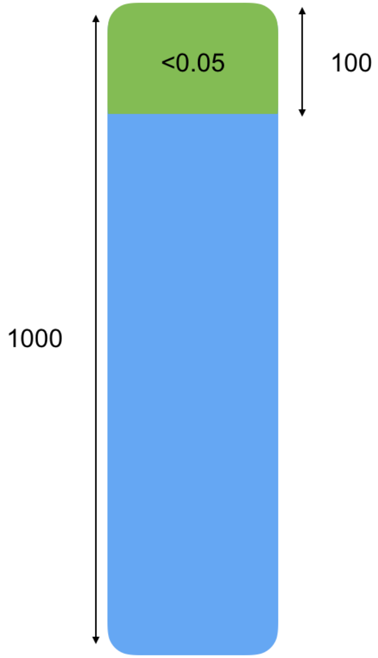
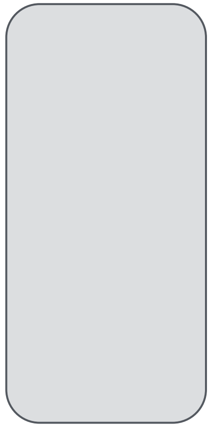




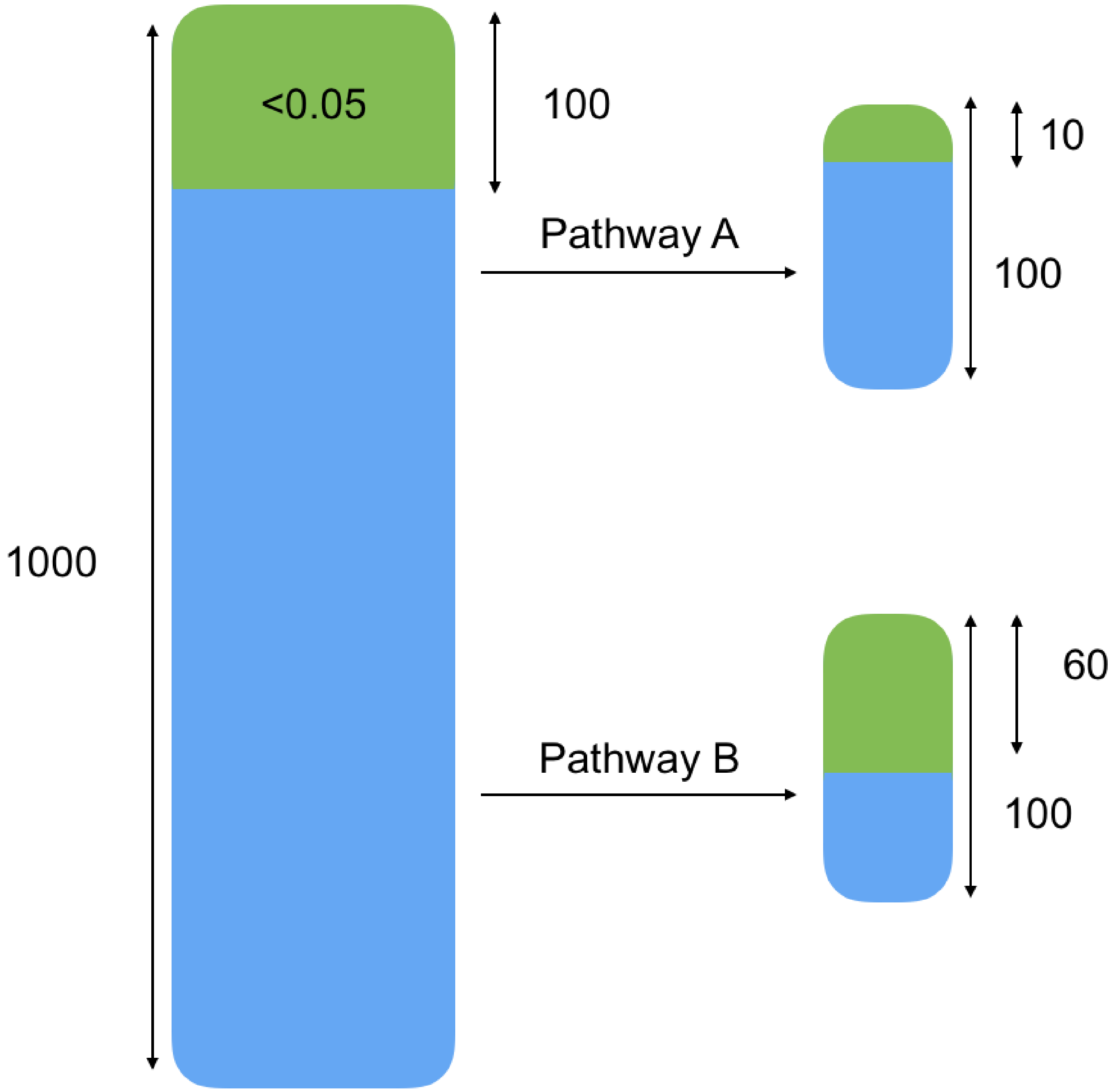
Imagine picking randomly 100 genes.



The proportion of deregulated genes follows a hypergeometric distribution.

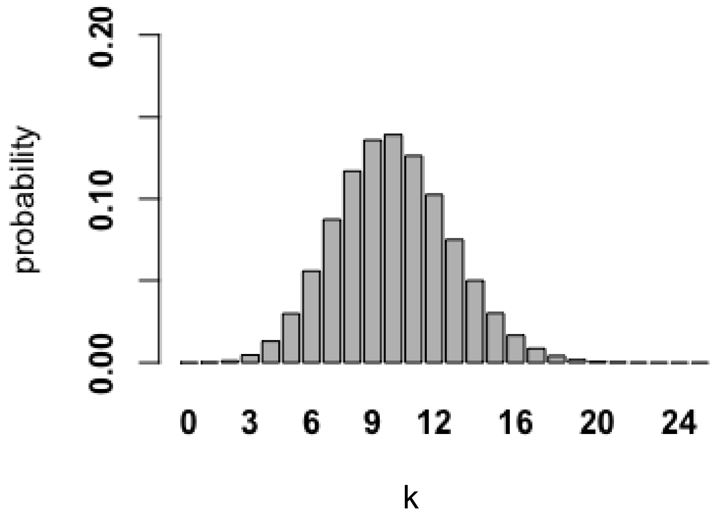
https://en.wikipedia.org/wiki/Hypergeometric_distribution#/media/File:HypergeometricPDF.png

Over-Representation Analysis (ORA)



- $N = 1000$
- $K = 100$

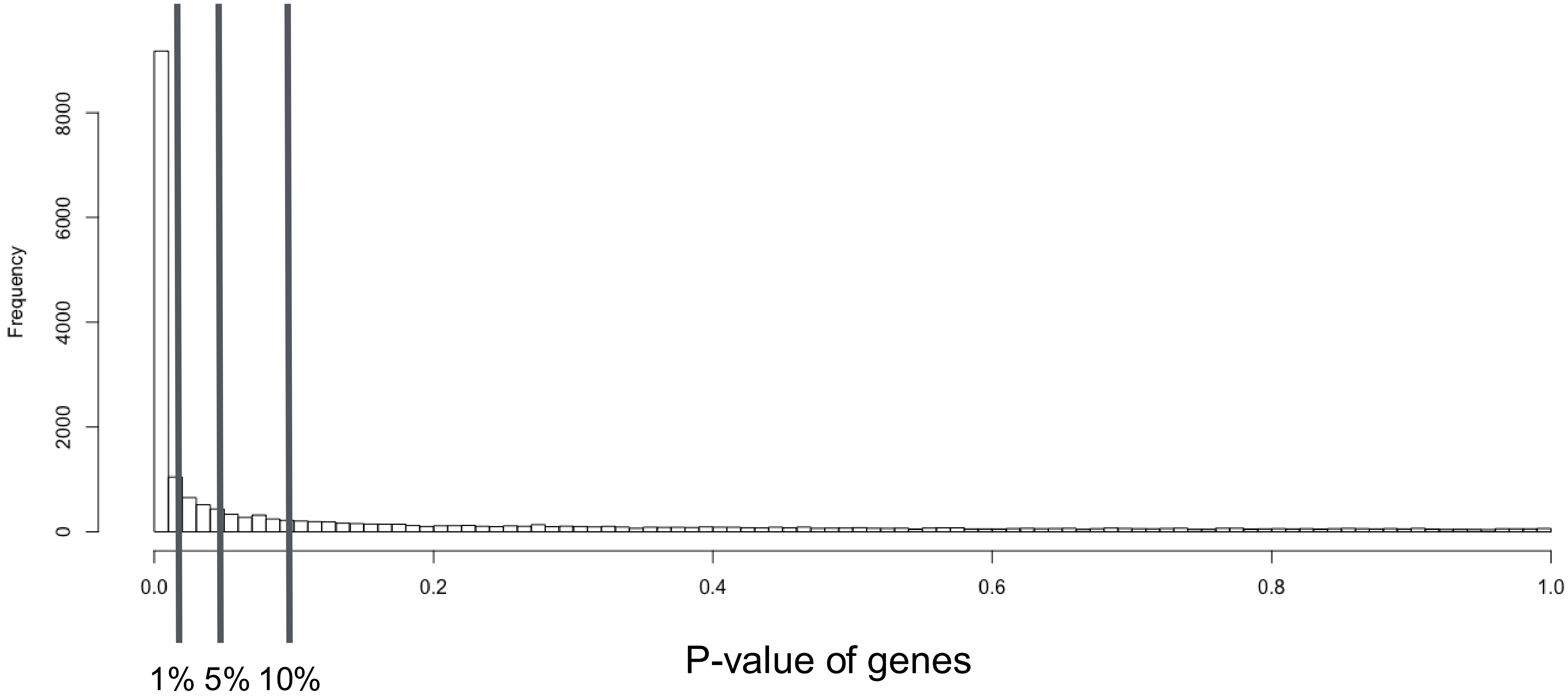
- $n = 100$
- $k = ?$



H_0 – proportion of genes in a group that are deregulated is similar to any other
H_1 – proportion of deregulated genes is significantly different to other groups

ORA – Major limitations

1) Where to put the threshold between deregulated/non-deregulated ?



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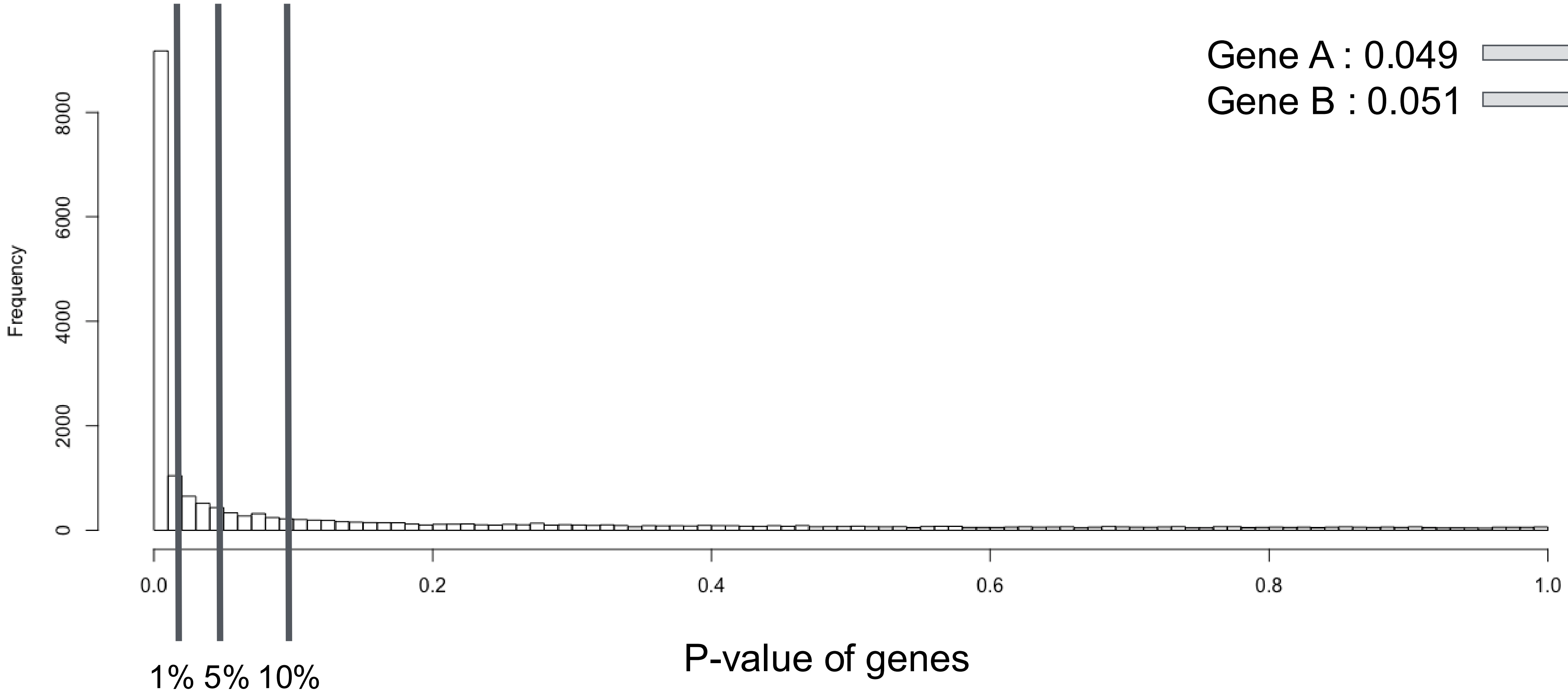

2) Over amplification of marginal differences.

Gene A : 0.049 → Deregulated

Gene B : 0.051 → Non-deregulated

ORA – Major limitations

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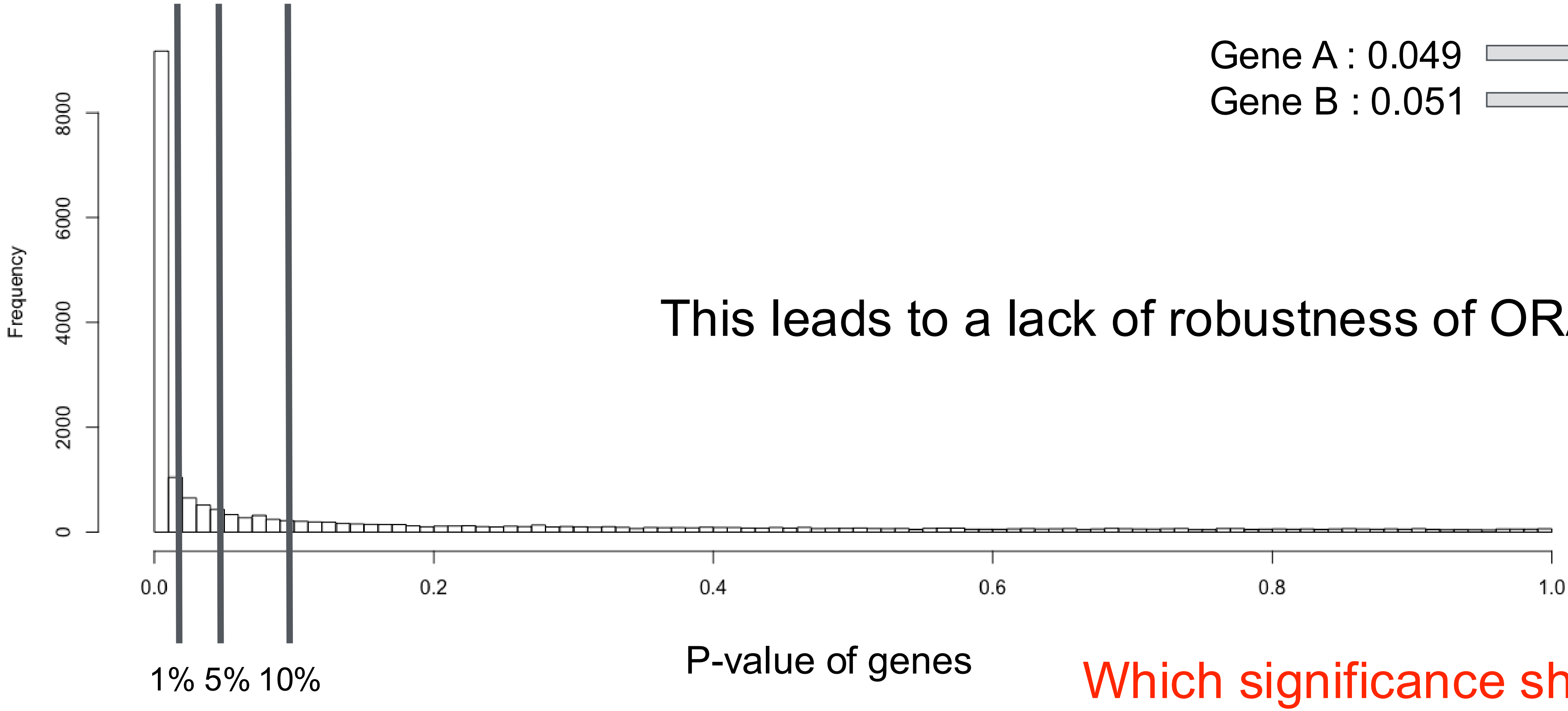


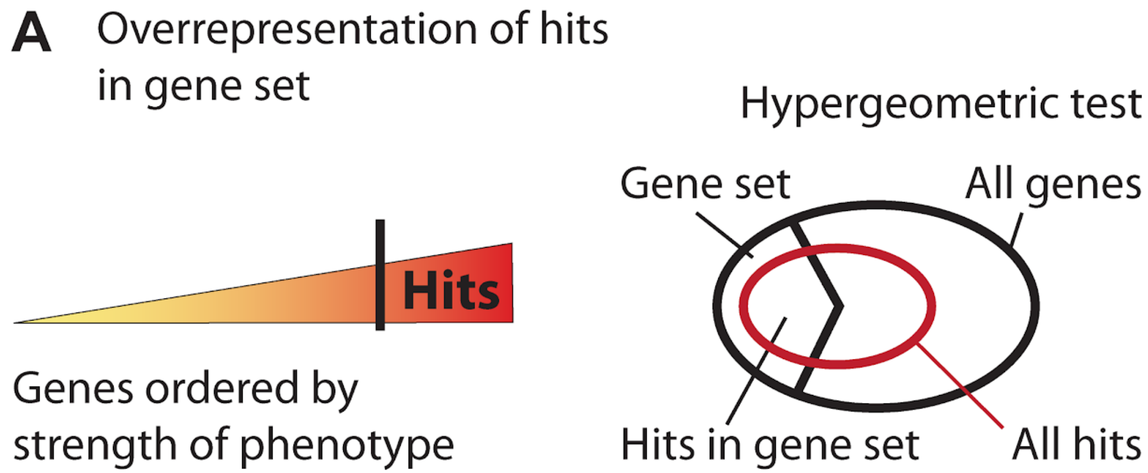
This leads to a lack of robustness of ORA

Which significance should we use?

Statistical enrichment analysis (SEA) addresses ORA limitations

Over-representation analysis

When comparing genes differentially expressed between two conditions, are there sets of genes that contain significantly more differentially expressed genes than expected?



Markowetz, Plos Comp Bio 2010

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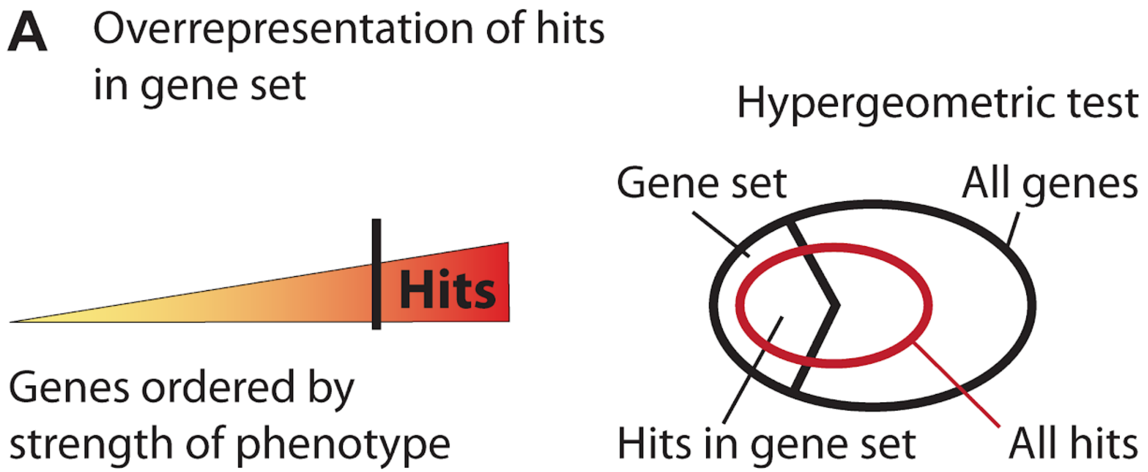
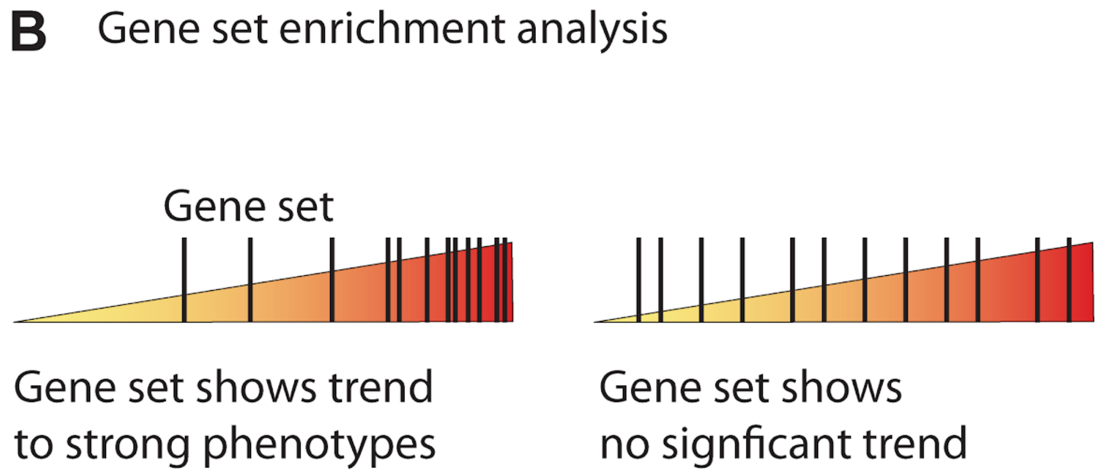


Statistical enrichment analysis

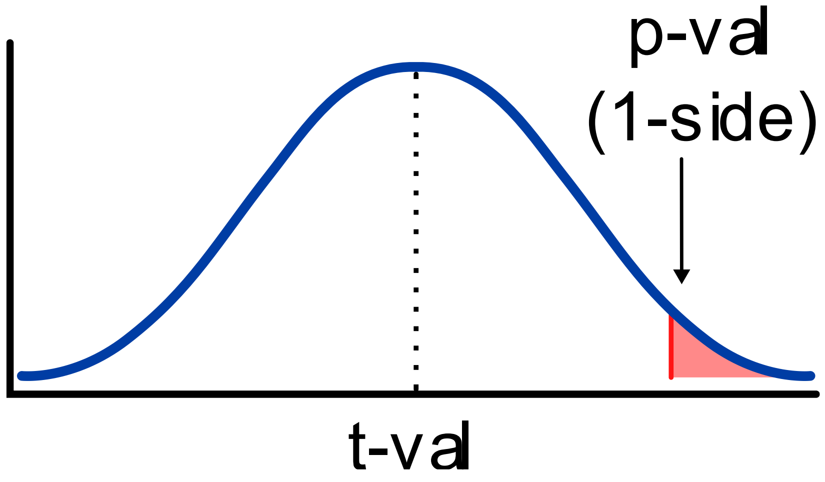
When comparing genes differentially expressed between two conditions, are there some sets in which the overall difference of expression is more extreme than expected?

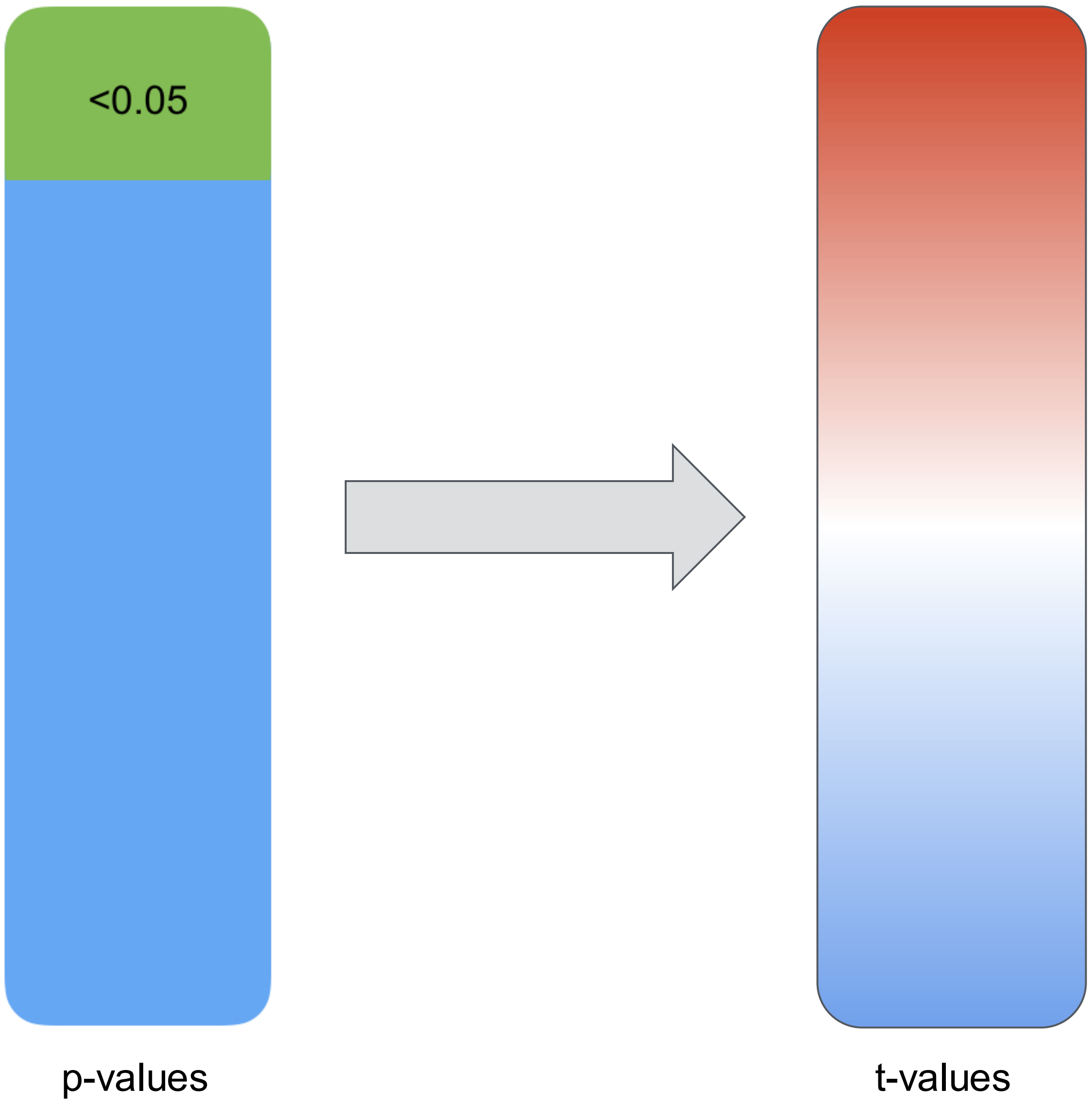


Markowetz, Plos Comp Bio 2010

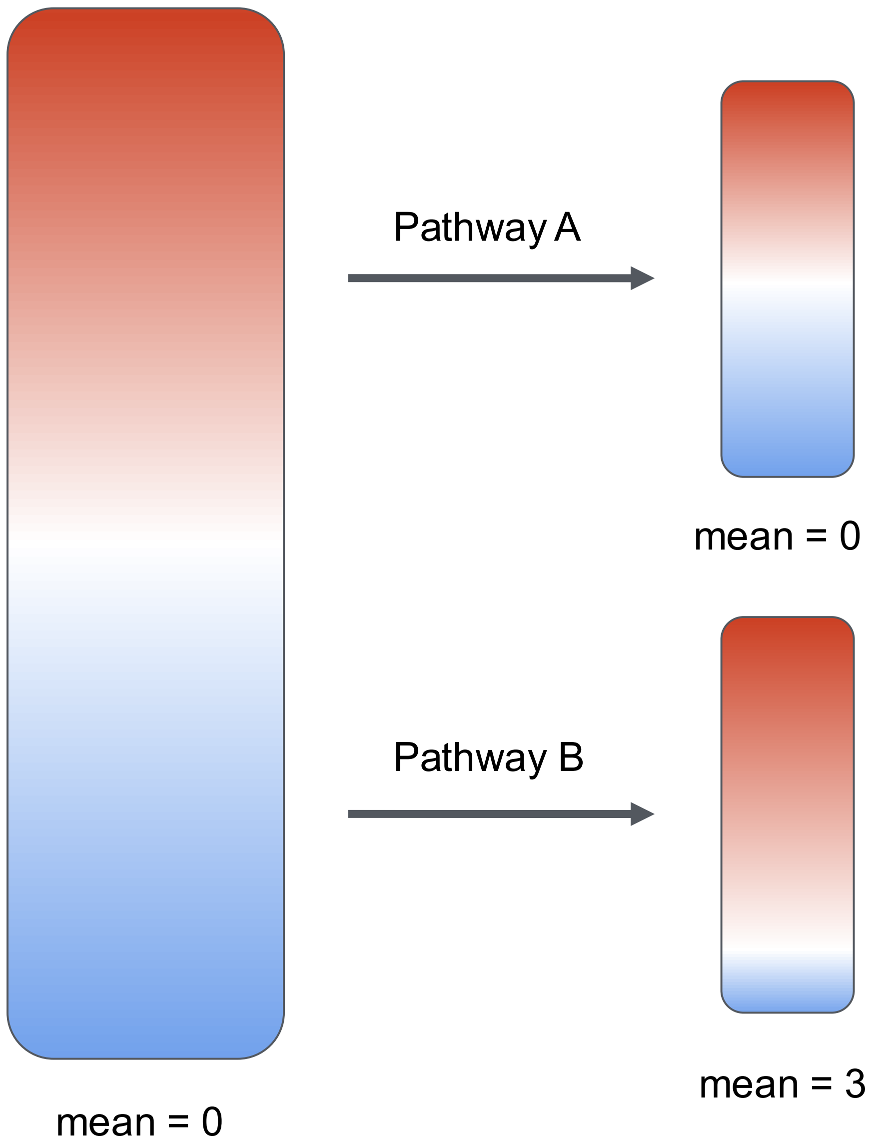
Statistical enrichment analysis (SEA)

- First we compute a metric (test statistic) e.g. t-value
- The corresponding p-value is the % of values of the test statistic distribution that are at least as extreme as the current t-value



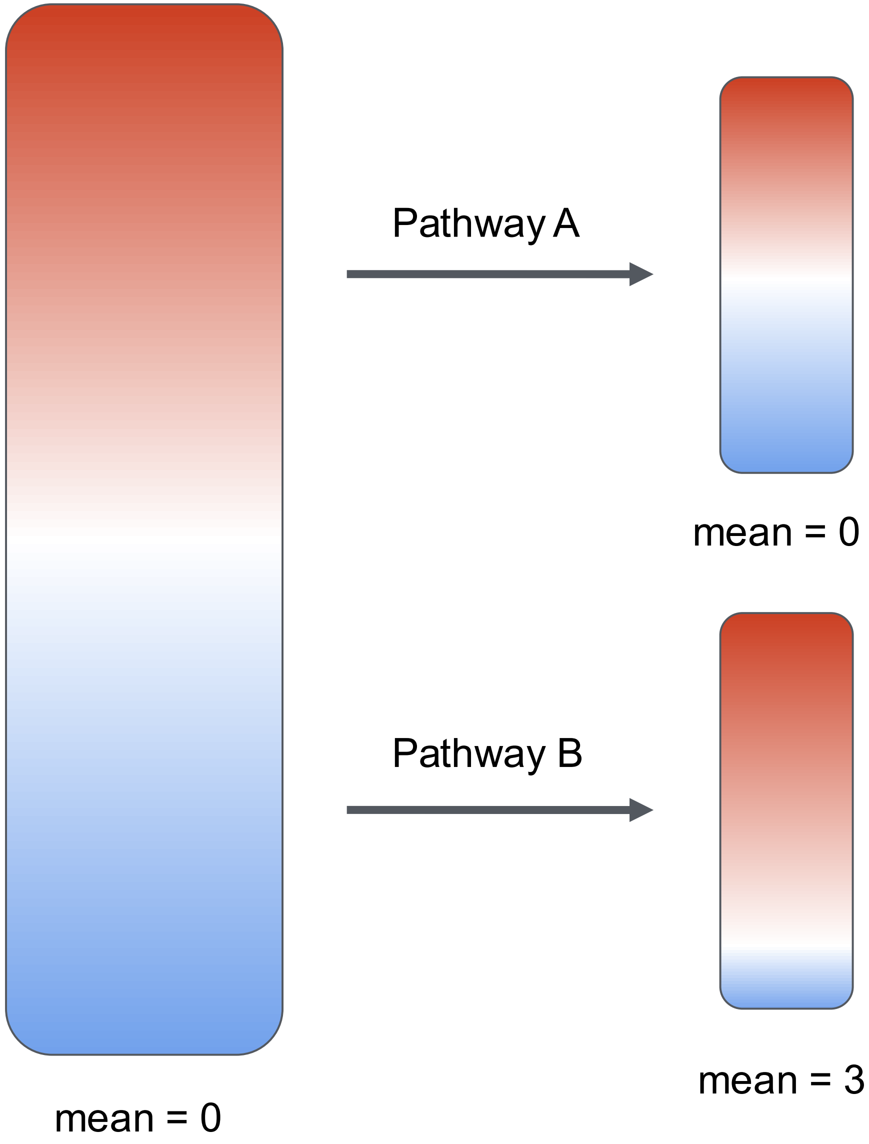


Statistical enrichment analysis (SEA)



Ideally the expected mean (set-level statistic) of all genes in a pathway is 0 if H_0 is true

Statistical enrichment analysis (SEA)

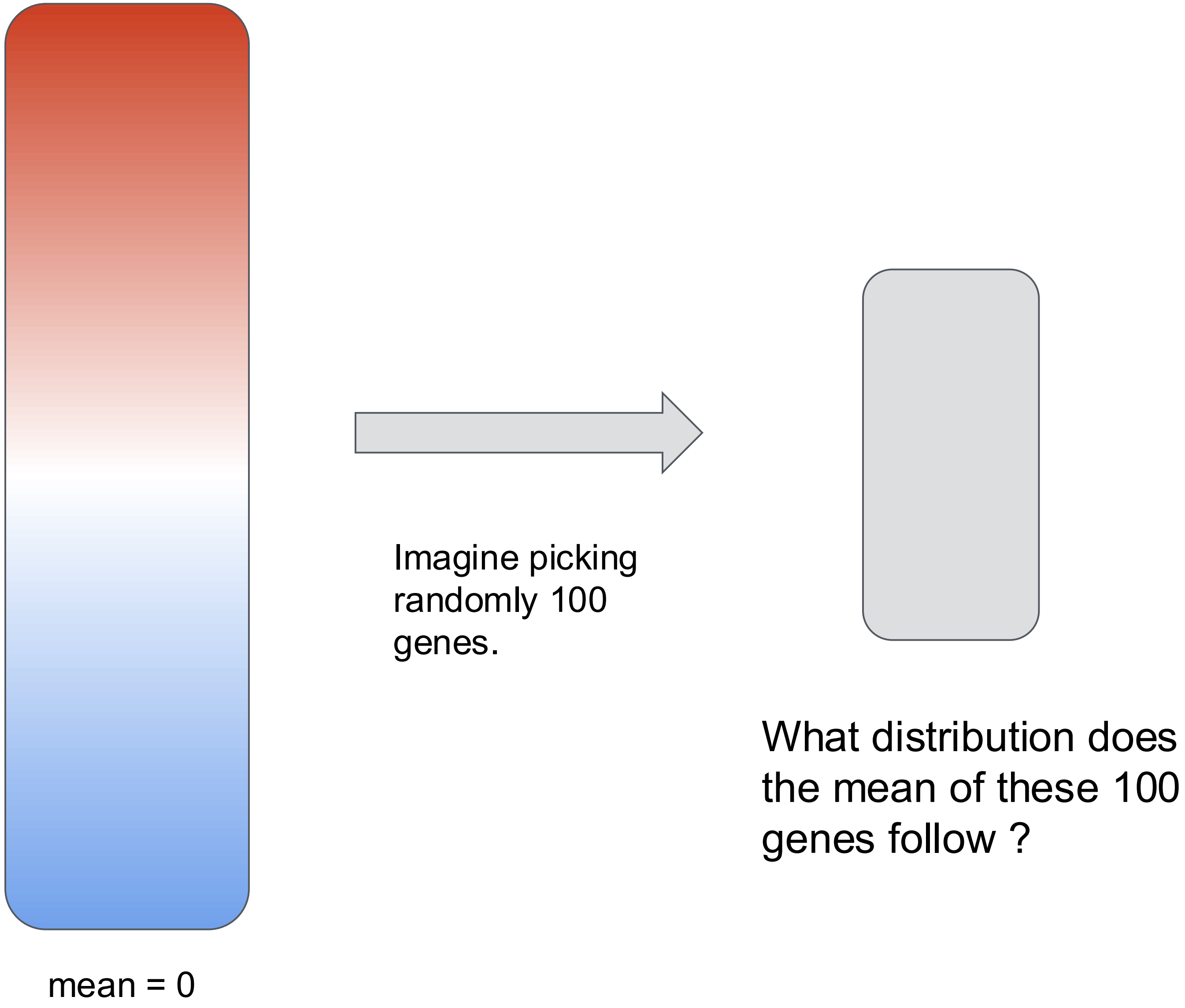


Ideally the expected mean (set-level statistic) of all genes in a pathway is 0 if H_0 is true

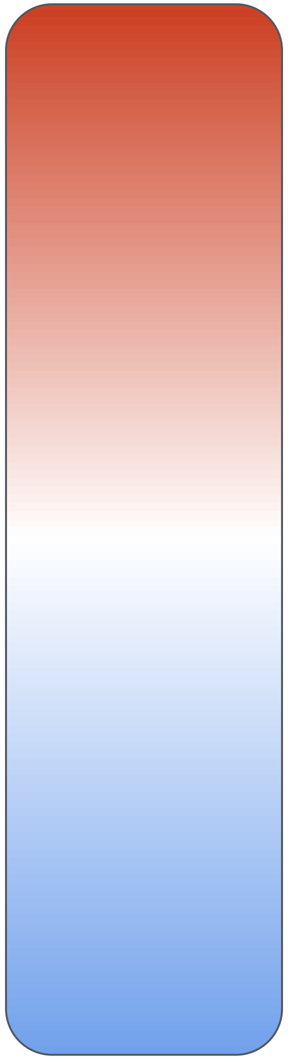
How likely is it to be so far from the expected mean by chance?

- Unknown distribution, so no analytical solution
- We use permutations to build a null distribution
- There are tools that do this for you :)

Empirical distributions: build your own distribution



Empirical distributions: build your own distribution



mean = 0

Imagine picking randomly 100 genes.

What distribution does the mean of these 100 genes follow ?

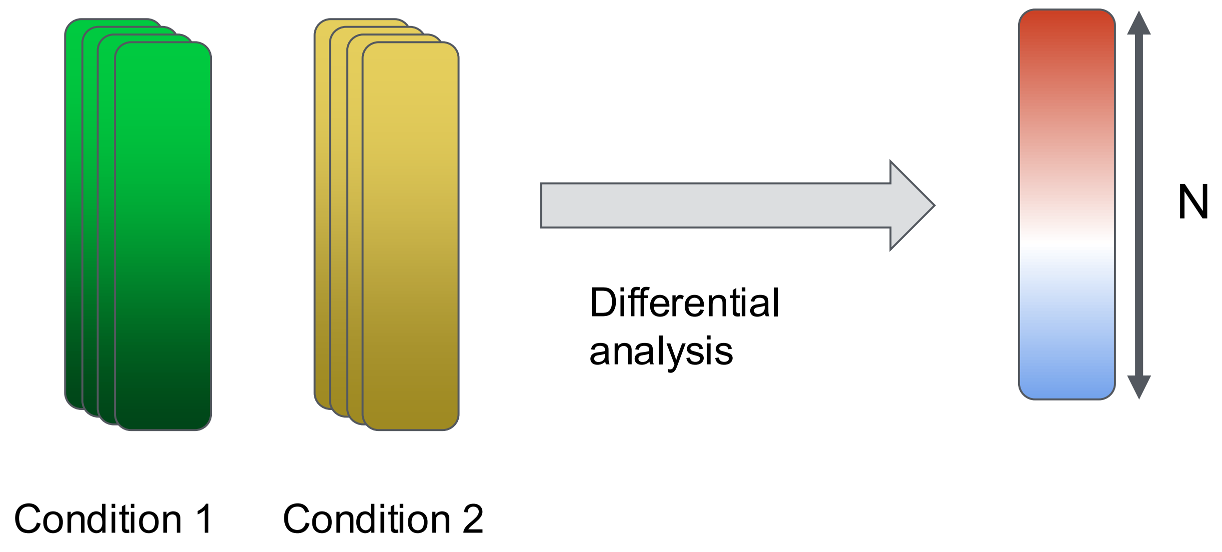
We are agnostic regarding the population distribution and create an *empirical distribution through permutation*.

Gene permutations

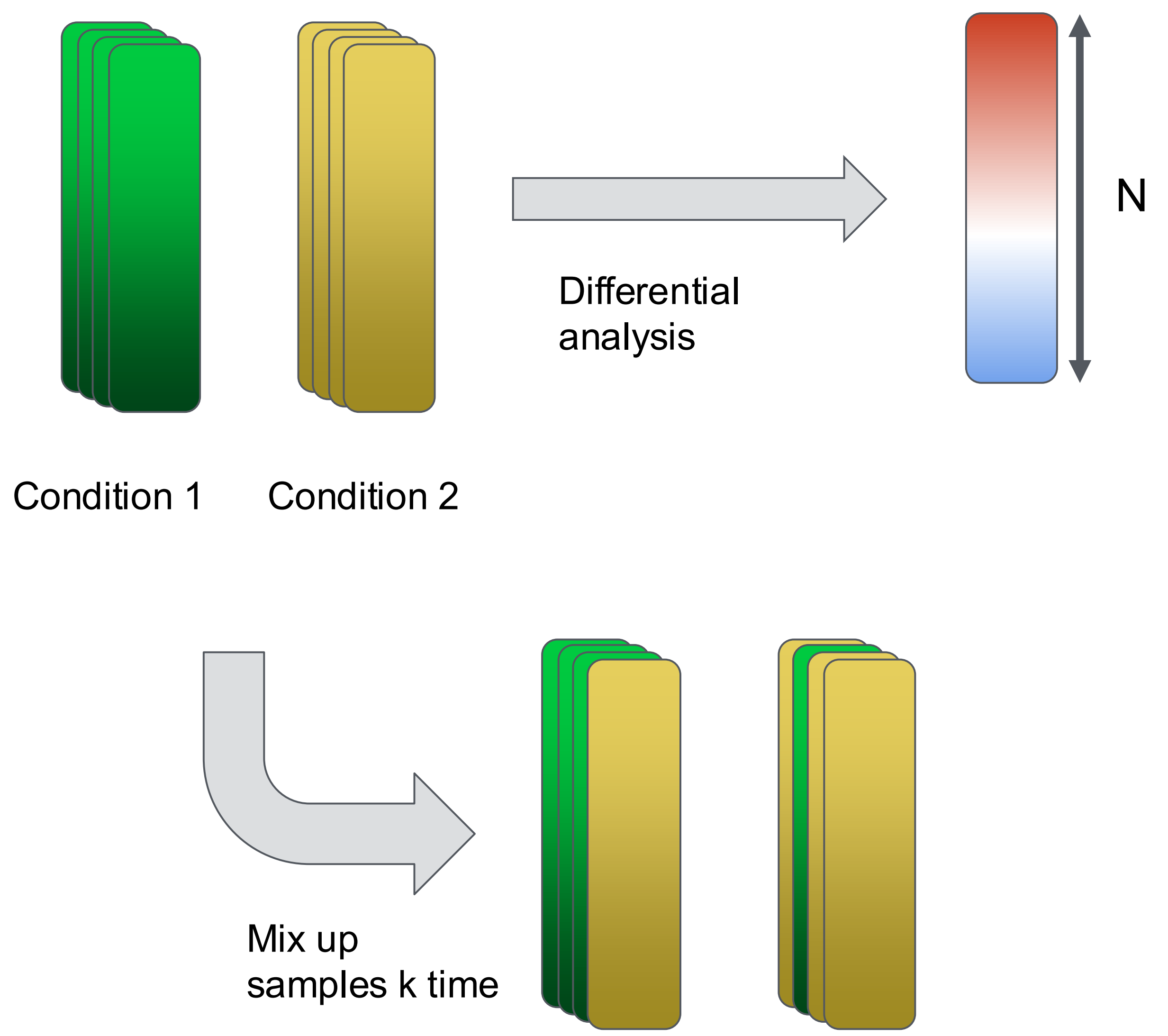
Sample permutations

See Tian et al. 2005 PNAS on the use of t-test for SEA

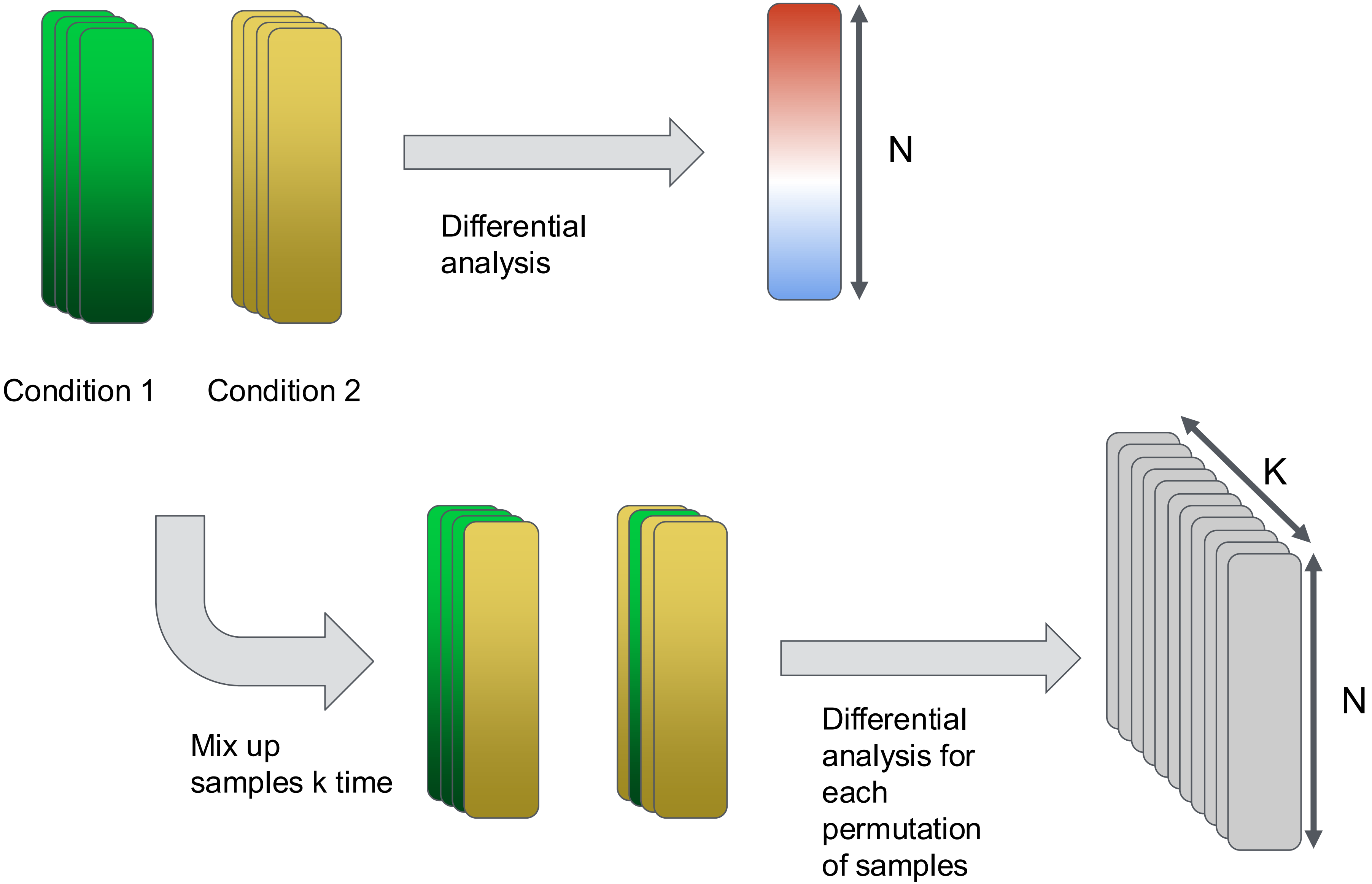
Empirical distribution by sample permutation



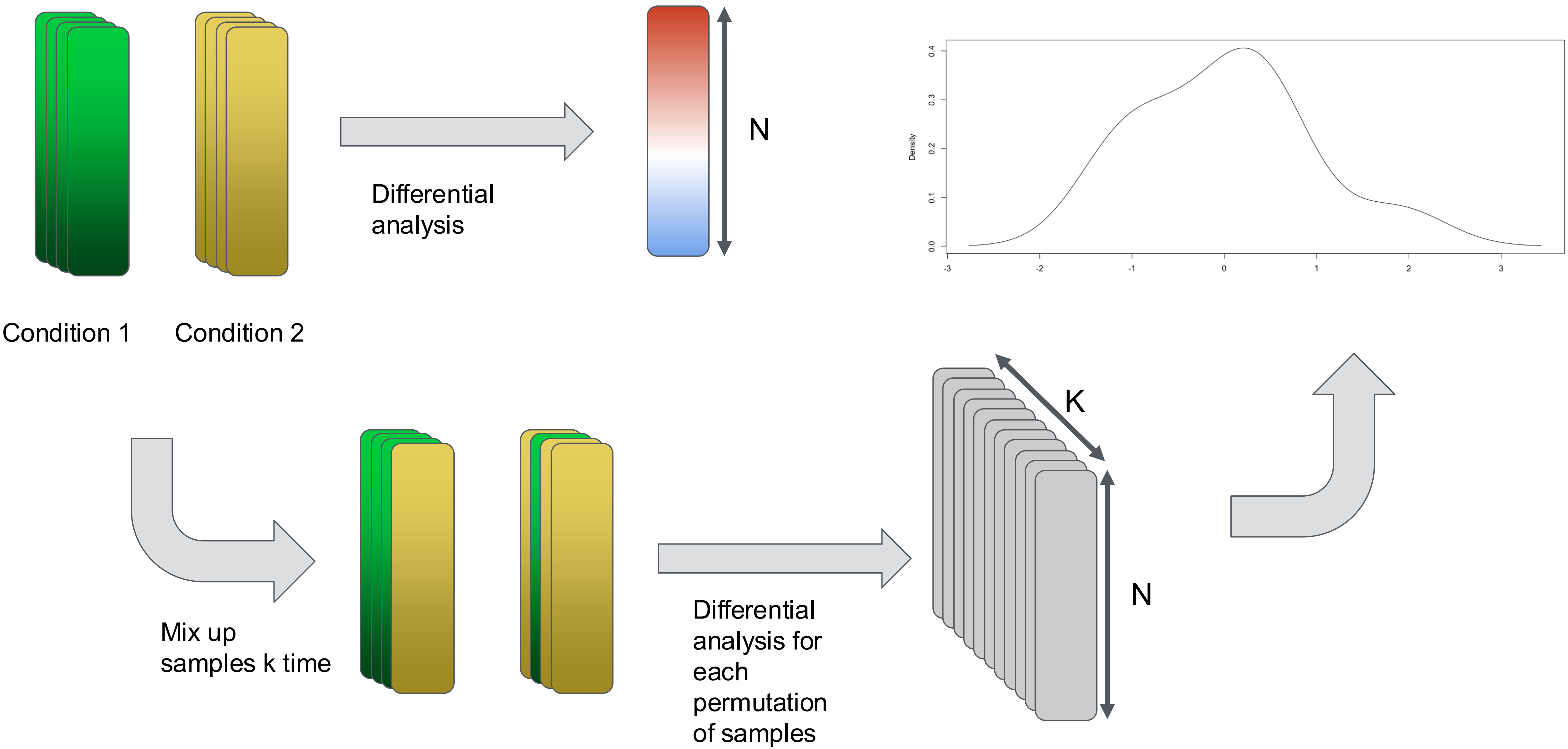
Empirical distribution by sample permutation



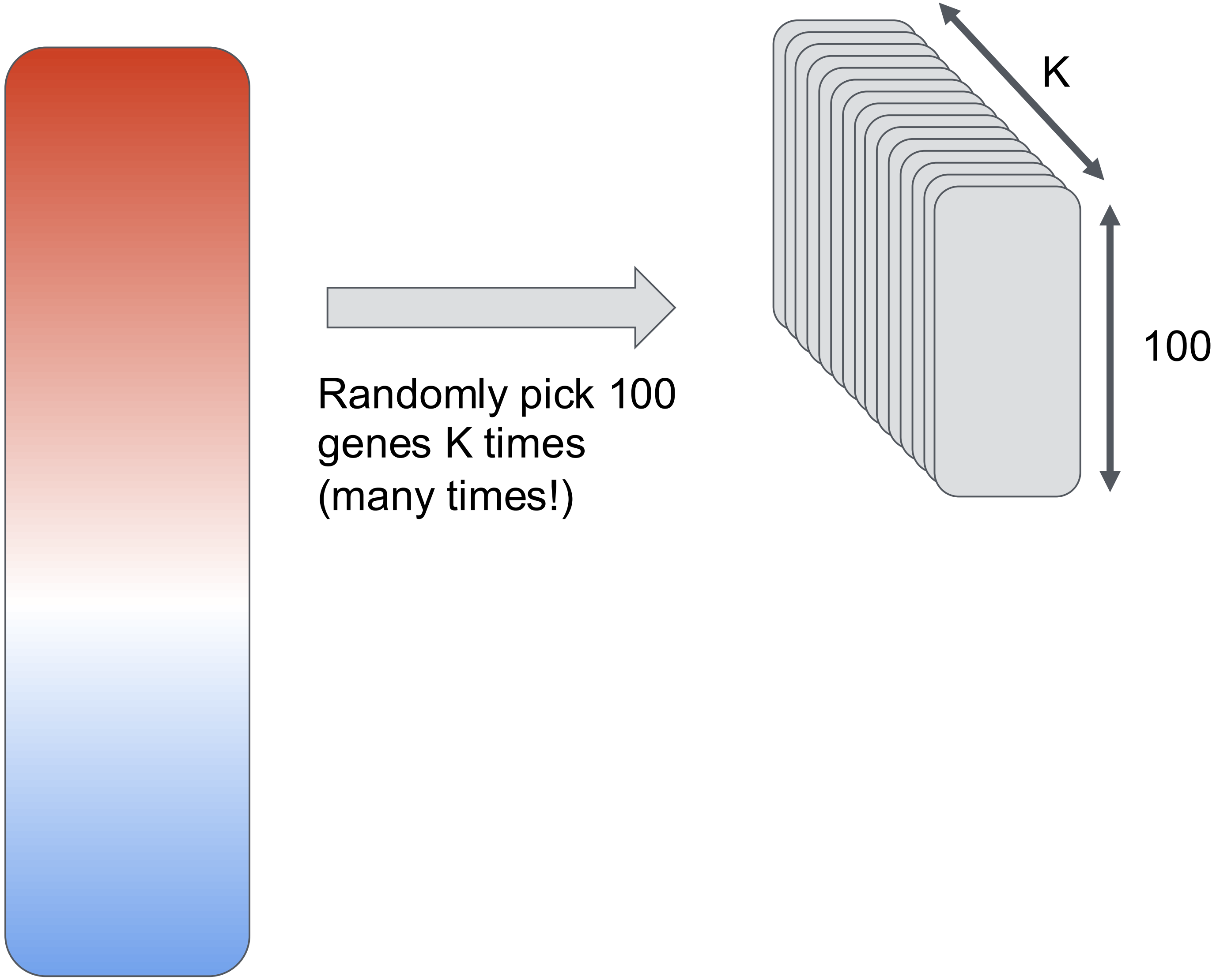
Empirical distribution by sample permutation



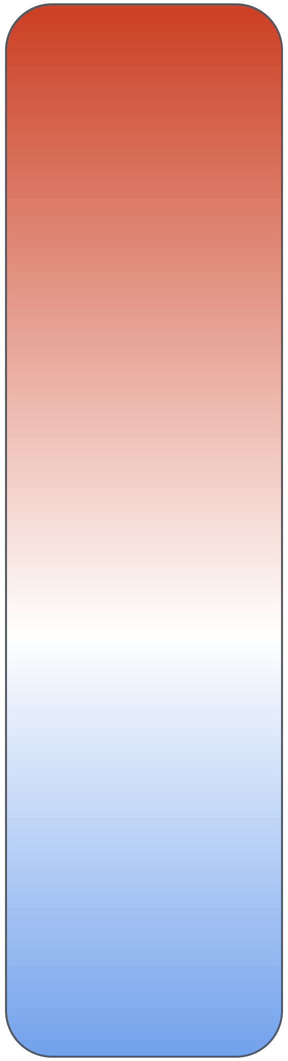
Empirical distribution by sample permutation



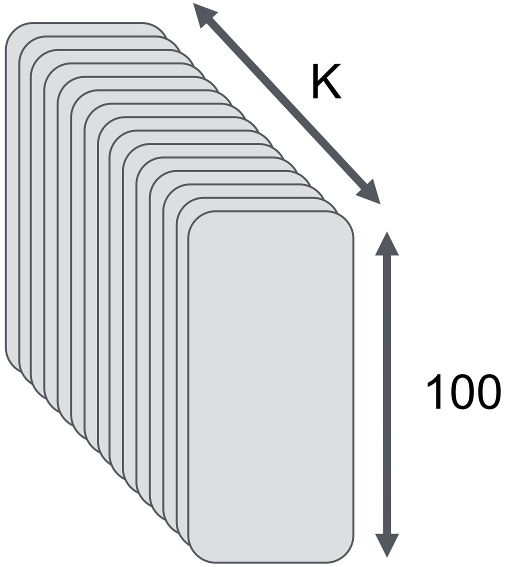
Empirical distribution by gene permutation



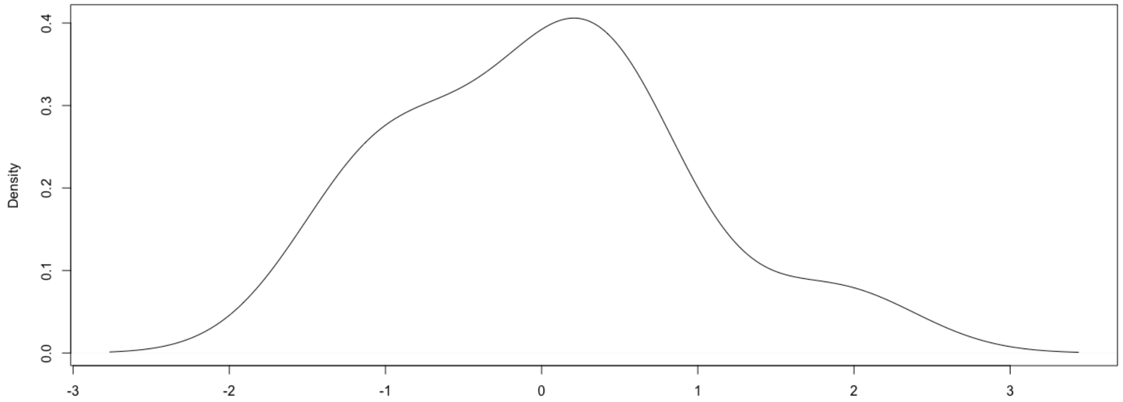
Empirical distribution by gene permutation



Randomly pick 100 genes K times (many times!)



Generate the distribution of means for each relevant n genes



Pros and cons of gene and sample permutation

Sample permutation :

Pros :

- Conserve the underlying correlation between members of pathways

Cons :

- Requires to repeat differential analysis multiple times so impossible to set up post hoc (need to rerun diff analysis)
- Computationally less efficient
- Requires a large number of samples

Gene permutation :

Pros :

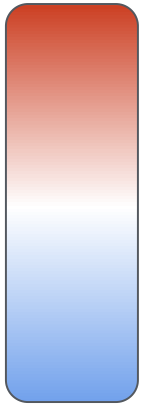
- Easy to set up post hoc
- Computationally efficient
- Can be done with any number of samples

Cons :

- Ignore the underlying correlation between members of pathways

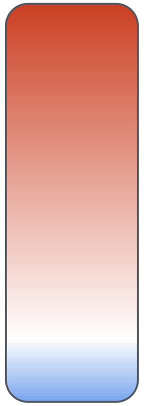
Compare genes in pathway against empirical distribution

Pathway A
(100 genes)



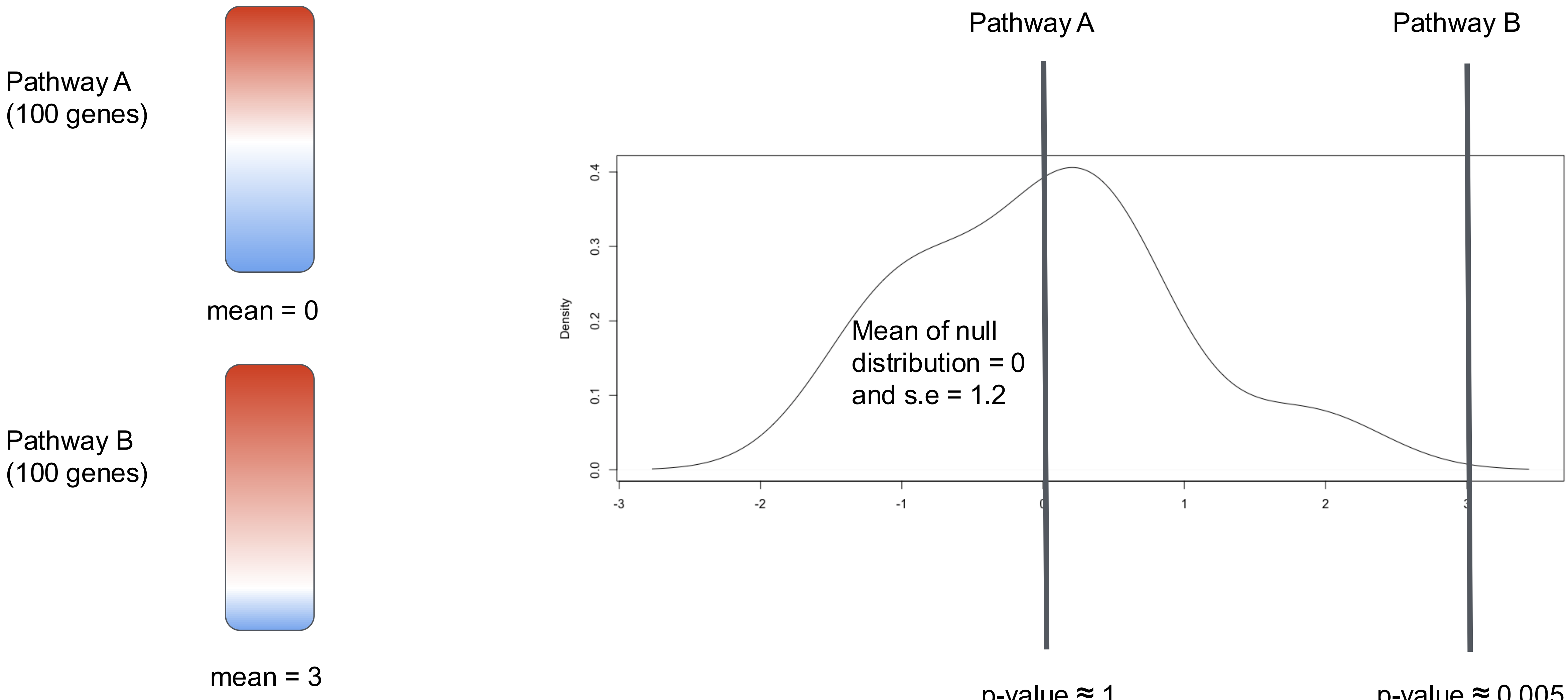
mean = 0

Pathway B
(100 genes)

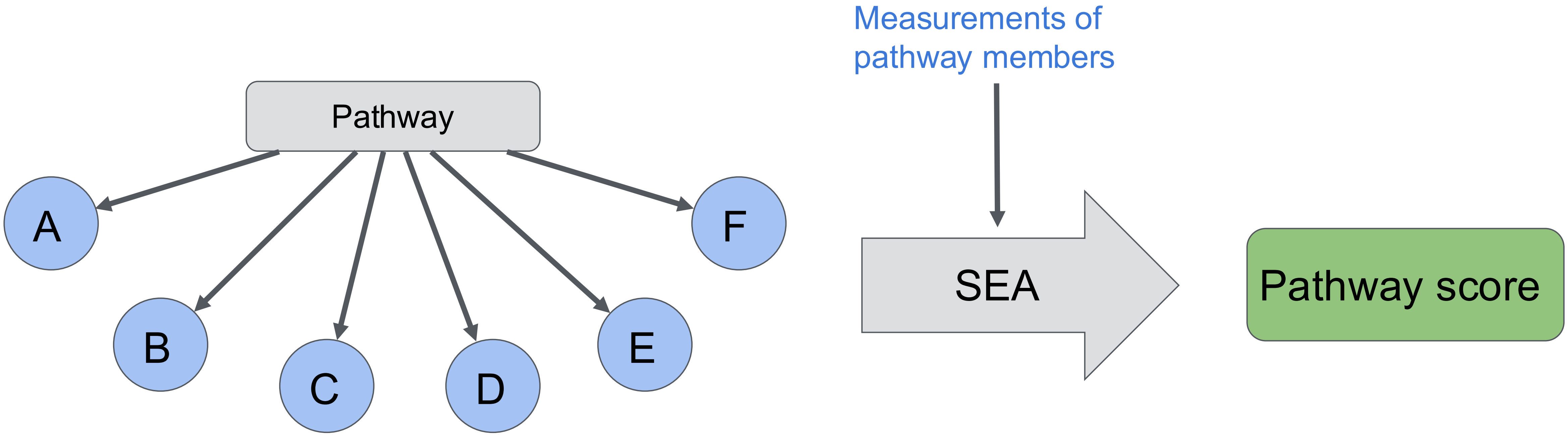


mean = 3

Compare genes in pathway against empirical distribution



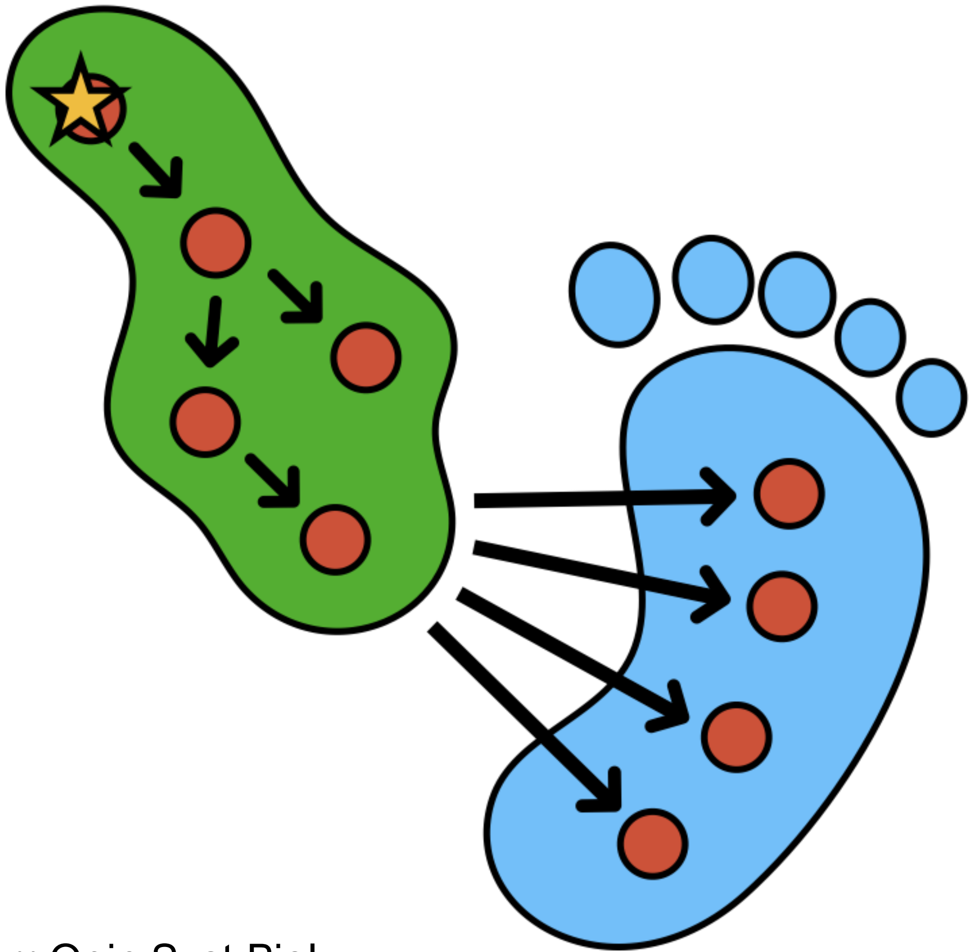
A pathway set can be used with SEA to estimate a pathway score



Score summarizes measurements of pathway members.
If measurements are transcript abundances, the score represent the overall abundance of transcripts of the genes in the pathway

Footprint based signatures

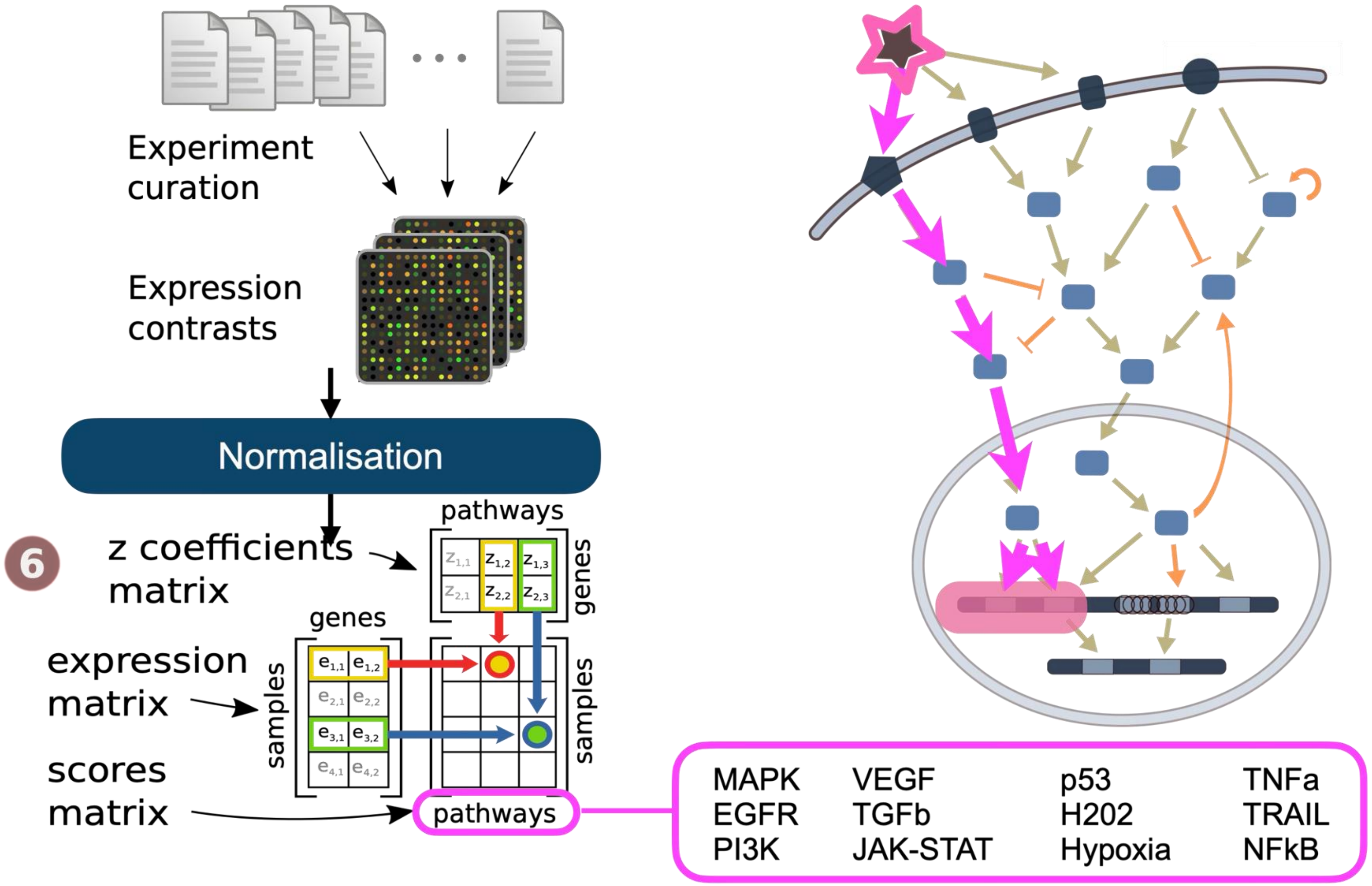
Direct information hard to access
activity of pathway



Indirect information
easier to access
transcriptomics

Dugourd A. and Saez-Rodriguez J. 2019. Curr Opin Syst Biol

Deriving mechanistic signatures from molecular data

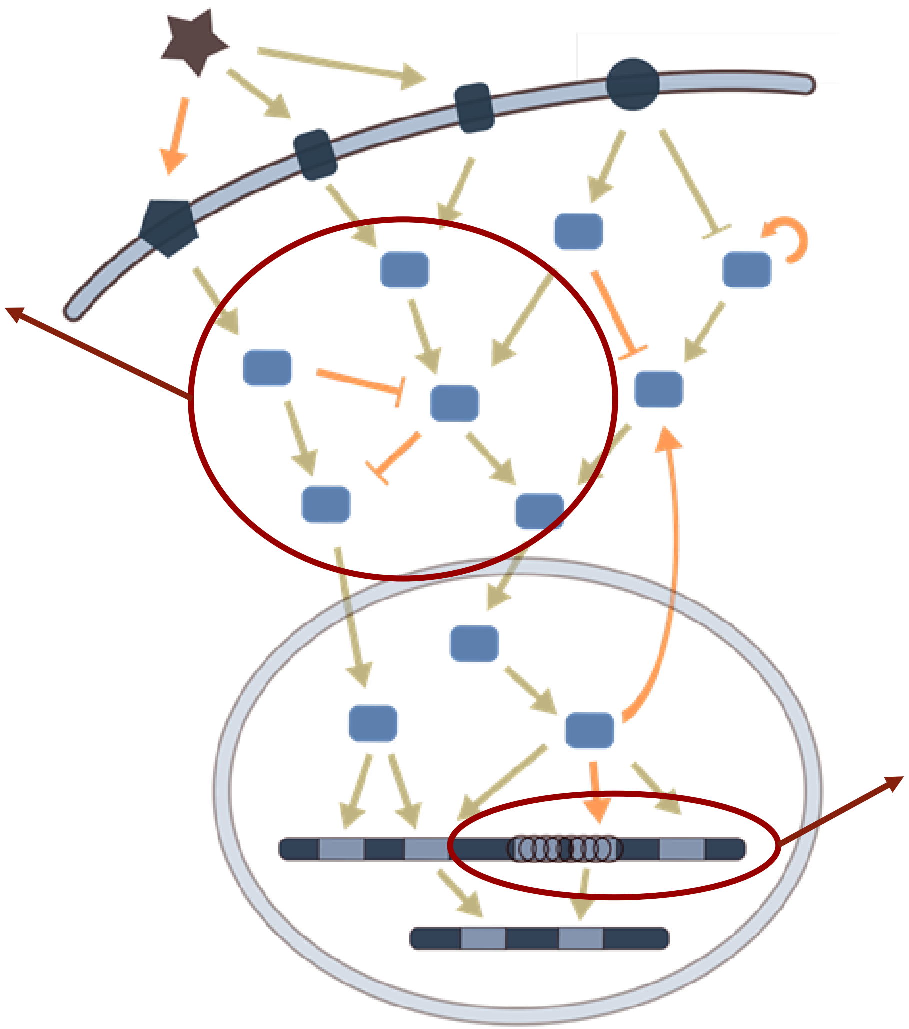


Schubert et al. 2018. Nature Communications

Feature sets vs footprints

Feature sets:

- Built based on the knowledge we have of a given biological process. E.g. kinases in a signaling pathway
- Included features might not be well measured in the omic used (e.g. kinases in transcriptomics data)

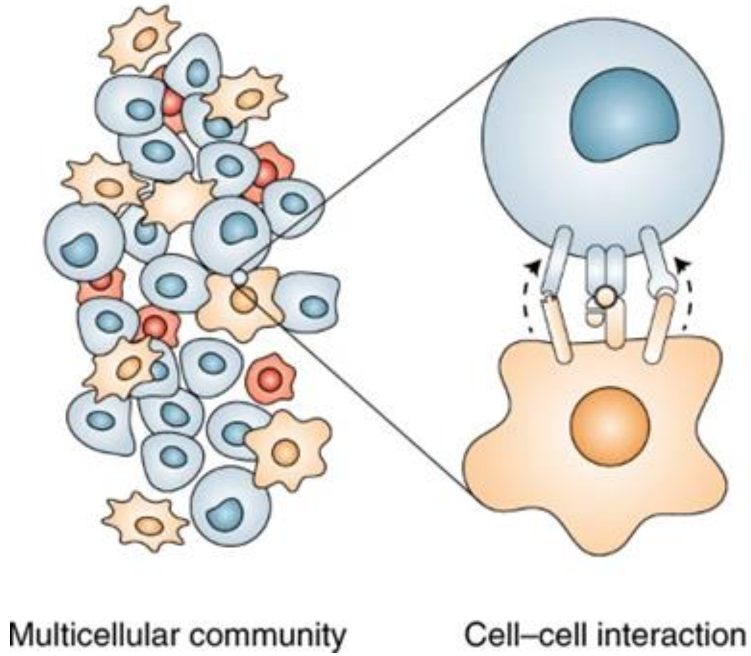


Footprints:

- Built based on the effect after a perturbation assay, “footprint”.
- Included features are well measured in the omic used (e.g. downstream genes after cell signaling)

Cell-cell communication

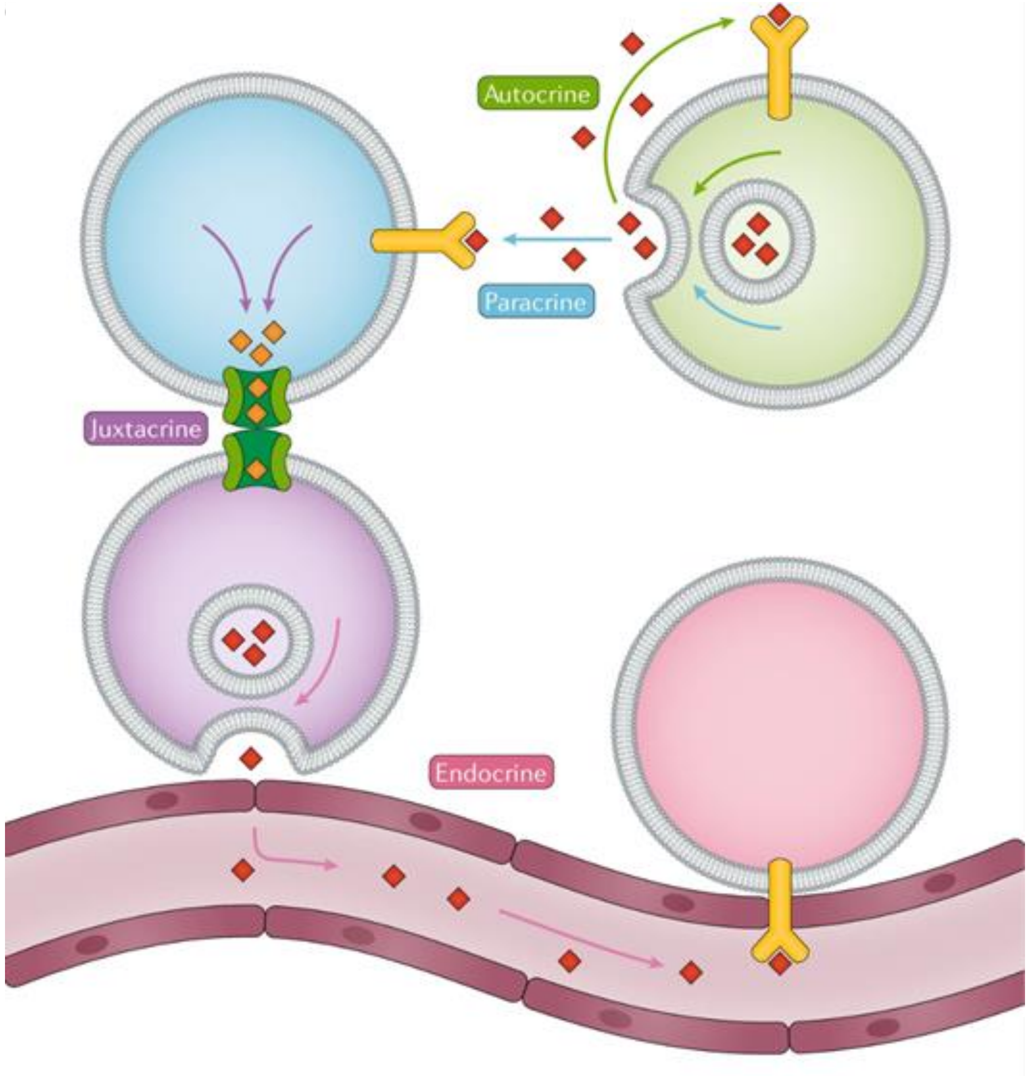
Cell signaling



- Cells are organized in communities
- Communicate in response to changes in their environment
- **Cell-Cell Communication** (CCC) via the interaction between proteins or molecules
- **Source** cell has a **ligand** that interacts with a **receptor** in the **target** cell

Bechtel T.J. *et al*. 2021. Nat Chem Biology

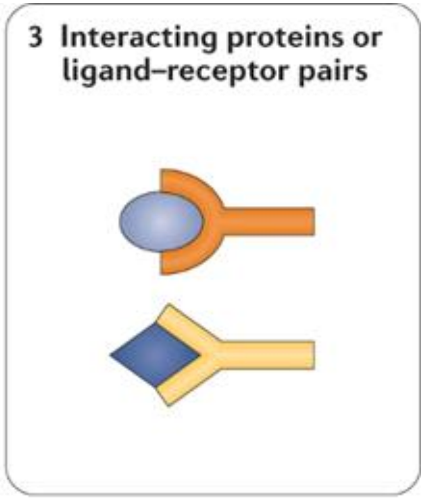
Types of cell signaling



- **Autocrine**: intracellular communication
- **Paracrine**: diffusion of molecules after secretion.
- **Juxtacrine**: contact-dependent without secretion
- **Endocrine**: long distance through fluids

Armingol E. *et al*. 2020. Nat Rev Genetics.

Prior knowledge of Ligand-Receptor pairs



Primary sources of Ligand-Receptor interactions:

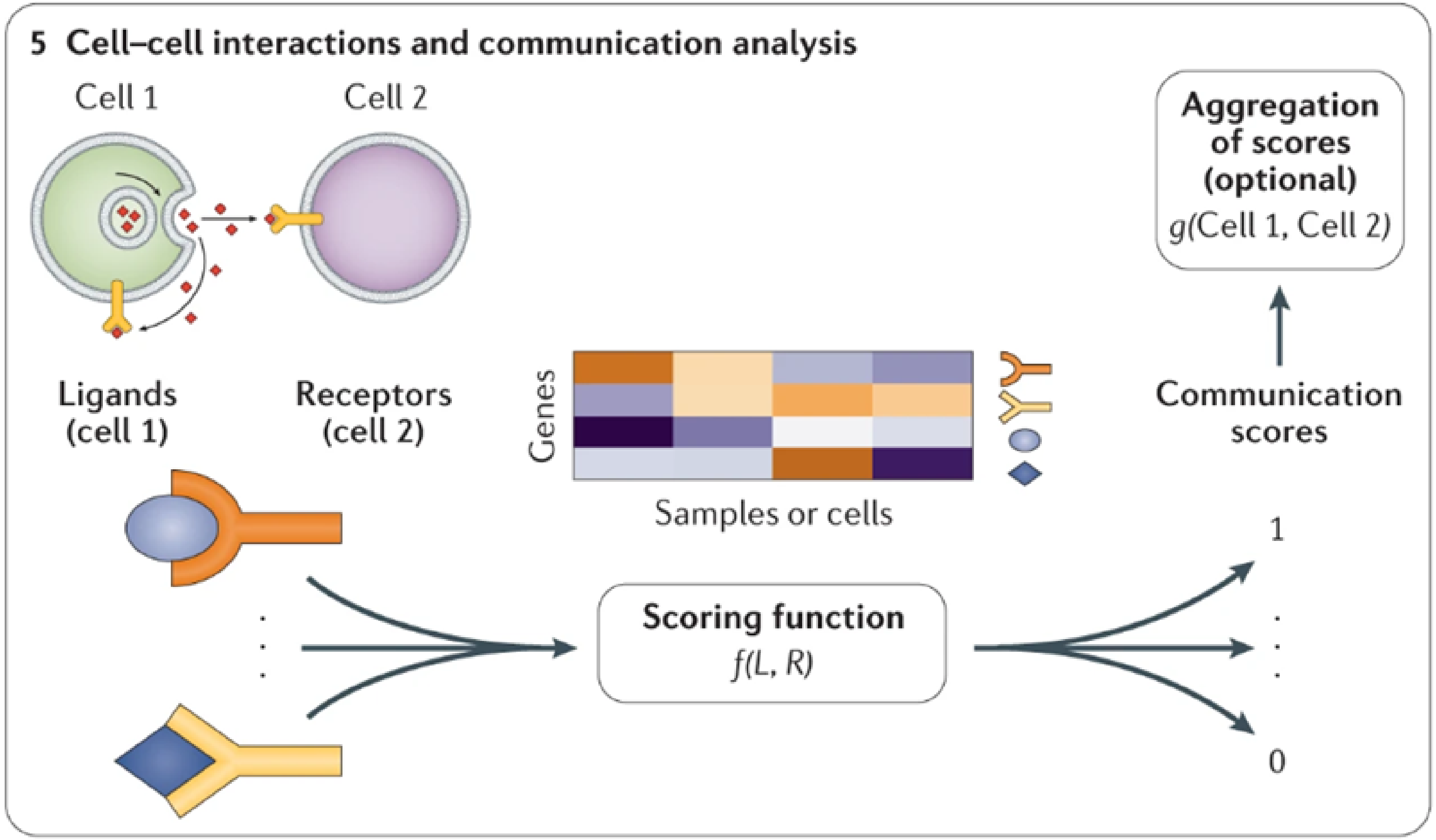
- Protein-Protein interactions (STRING, IntAct, …)
- Annotated in gene sets (REACTOME, KEGG, …)
- Manual curation from literature (FANTOM5, HMPR, …)

Meta-resources:

- CellPhoneDB
- iTALK
- connectomeDB
- OmniPath
- …

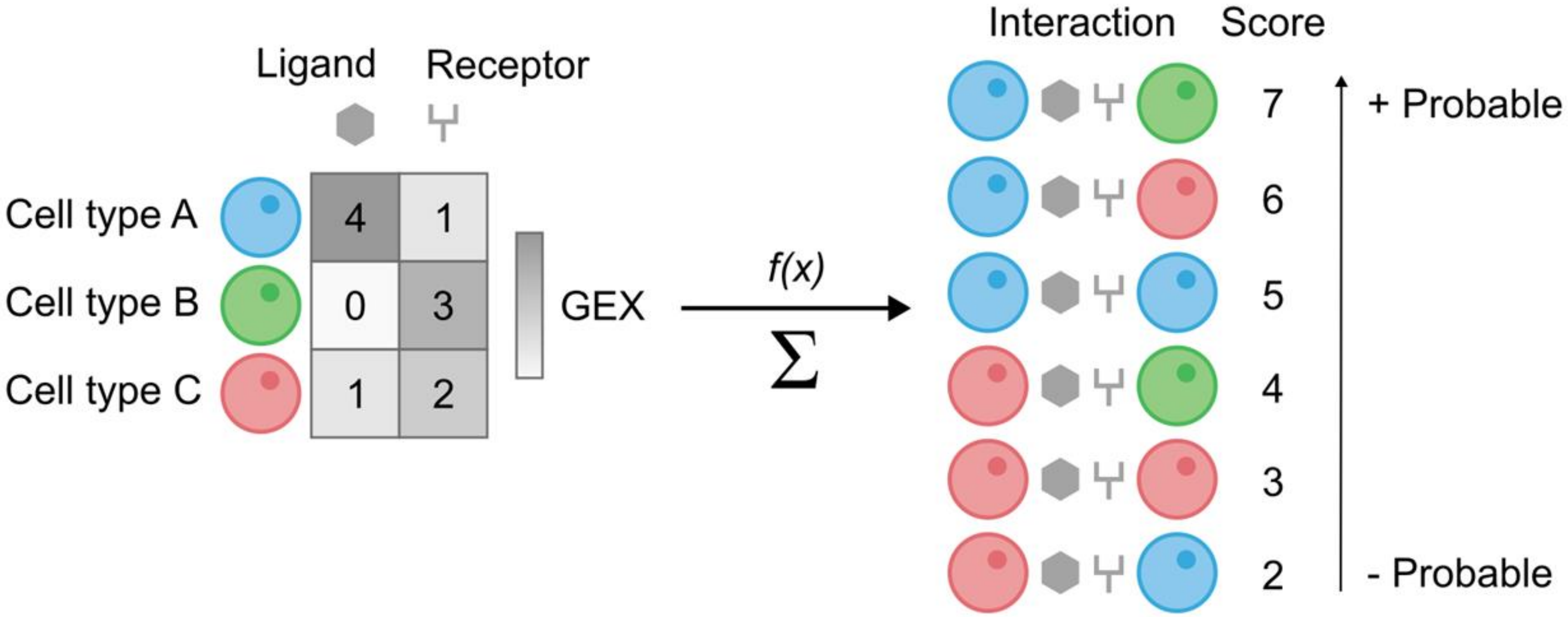
Armingol E. *et al*. 2020. Nat Rev Genetics.

Filering by prior knowledge



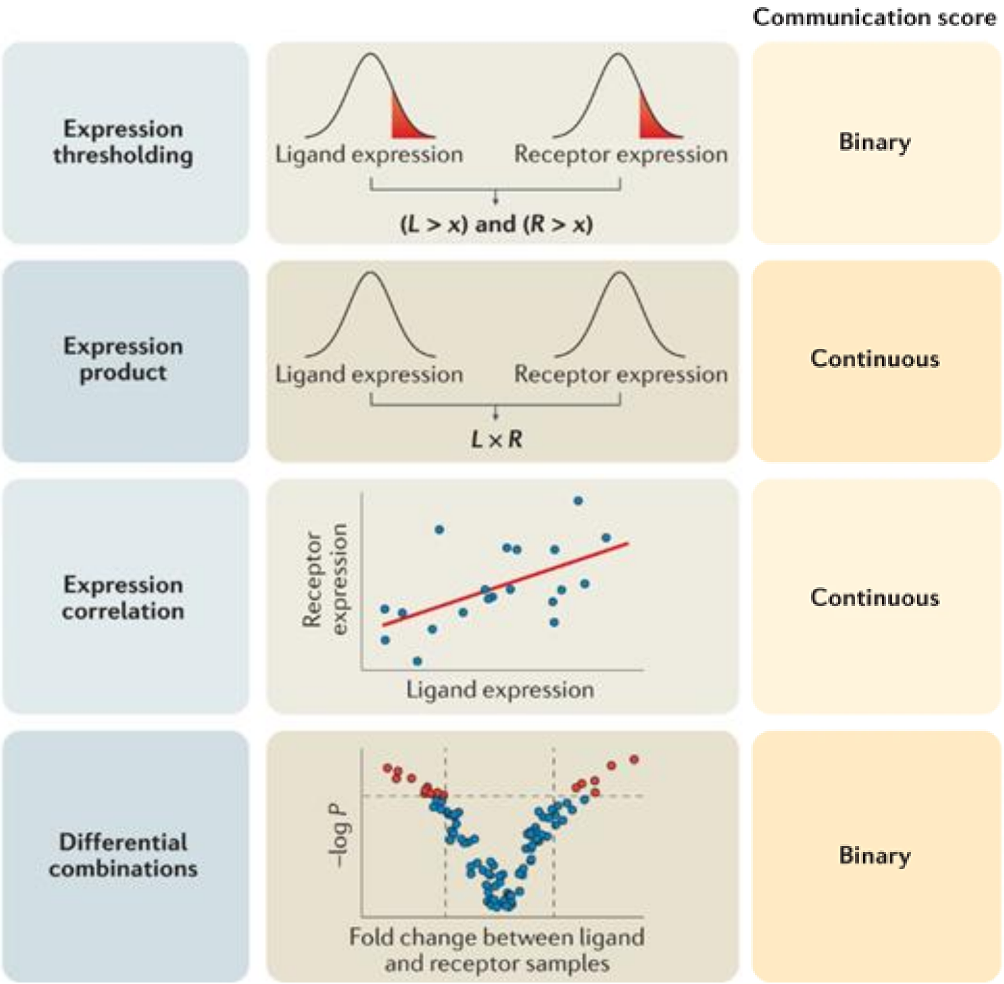
Armingol E. *et al*. 2020. Nat Rev Genetics.

Toy example



Ligand
Receptor
Cell type A
Cell type B
Cell type C
4
1
0
3
1
2
GEX
f(x)
Σ
Interaction
Score
7
6
5
4
3
2
+ Probable
- Probable

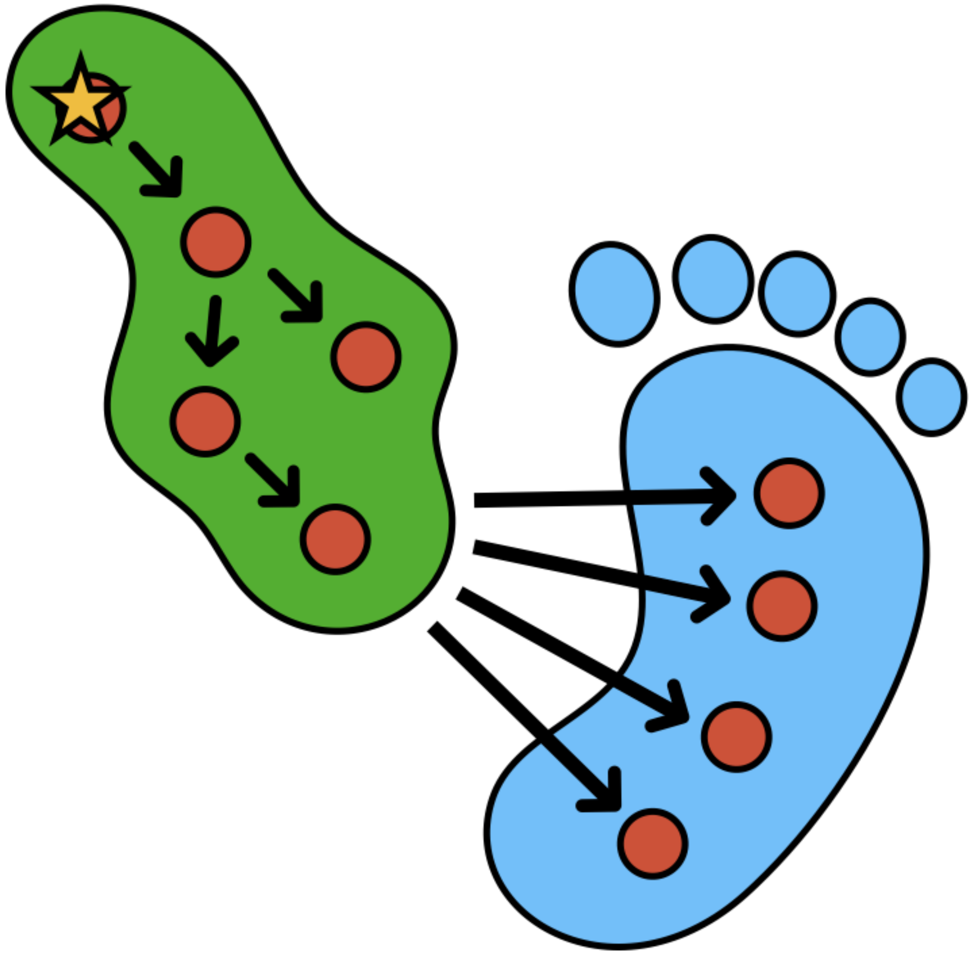
Strategies for scoring functions



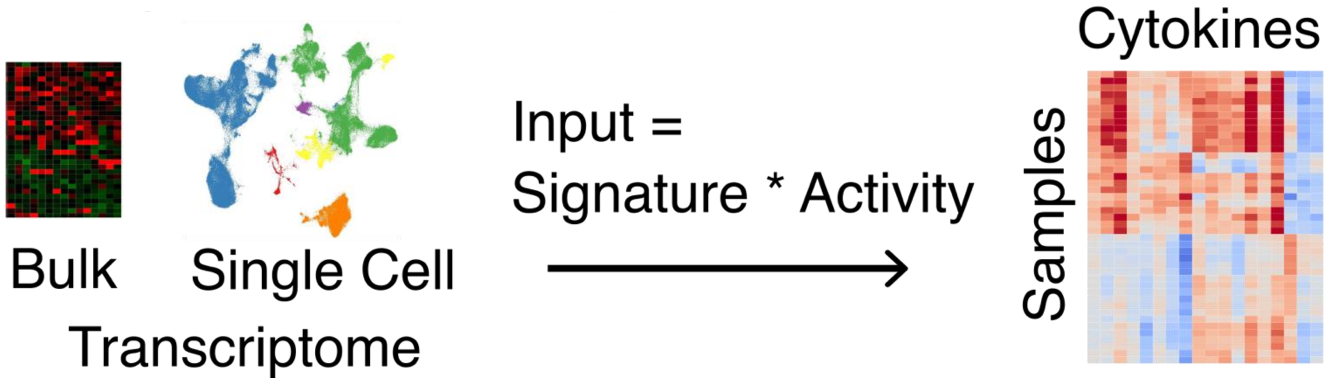
- **2 types of scores:**
  - Magnitude (strength)
  - Specificity (uniqueness across cell types)

Armingol E. *et al*. 2020. Nat Rev Genetics.

Footprint based analysis



CytoSig – cytokine activity
Jiang et al. 2021 Nature Methods



RIDDEN – receptor activity
Barsi et al. 2025 PLOS Comp Biol

Practical – Data exploration

Access one or more of the following data resources. Browse the collections and select an interesting dataset to work with. Mind the size of the data. The analyses should be run on your computer.

- HCA - https://data.humancellatlas.org
- CELLxGENE - https://cellxgene.cziscience.com/datasets
- HuBMAP - https://portal.hubmapconsortium.org
- HTA - https://humantumoratlas.org/explore

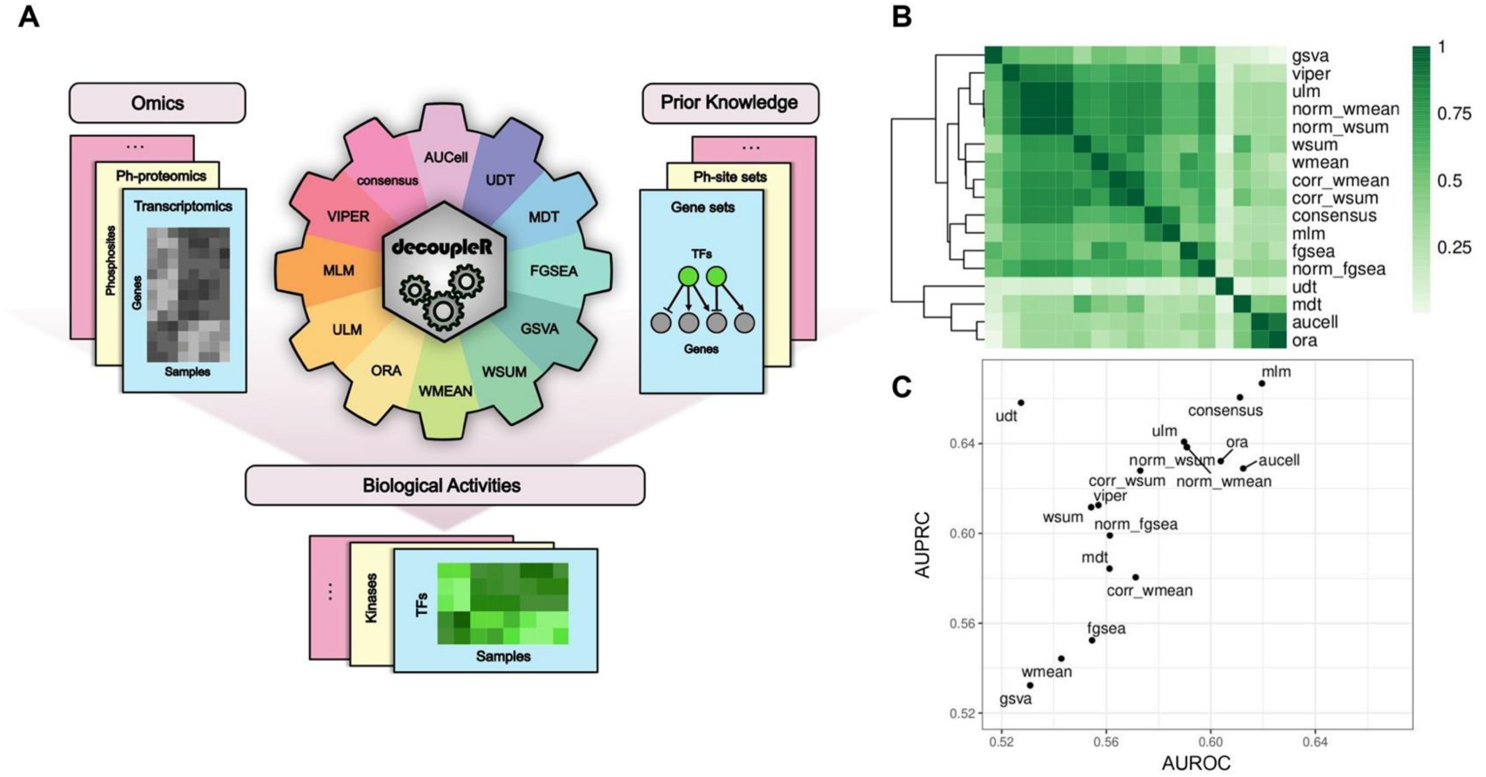
What is the experimental design?
Is the data normalized and variance stabilized?
If there are multiple conditions or batches is there a noticeable batch effect? Keep this in mind for downstream analyses.
When in doubt about the data use the pbmc3k dataset (raw: https://cf.10xgenomics.com/samples/cell/pbmc3k/pbmc3k_filtered_gene_bc_matrices.tar.gz, processed: scanpy.datasets.pbmc3k_processed)

To help you with the data exploration:
- If working in Python follow https://www.sc-best-practices.org using the anndata and scanpy packages
- If working in R follow https://bioconductor.org/books/3.21/OSCA/ for analysis using the Bioconductor ecosystem or https://satijalab.org/seurat/articles/get_started_v5_new for analysis with Seurat

Practical – Enrichment analysis

Enrichment analysis with decoupleR (https://decoupler.readthedocs.io, https://saezlab.github.io/decoupleR/)
Prior knowledge resources can be conveniently accessed through the Omnipath module in decoupler
Use the data from the previous exercise or switch. Does your data contain multiple conditions? Did you already perform differential expression analysis? Remember that you can also compute scores per sample.

Resources:
- MSigDB (H: hallmarks, C2: curated gene sets)
- PROGENy (cancer related pathways)
- PangalaoDB (cell type)
- CytoSig (cytokine activities)



Methods:
- ORA
- FGSEA
- ULM
- MLM
- Cell type enrichment (PanglaoDB)

Compare the findings.

Practical – Cell-cell communication

Enrichment analysis with LIANA(+). Use the same data as before and continue your analysis or switch to the PBMC data as described in the links below.

Python - Follow https://liana-py.readthedocs.io/en/latest/notebooks/basic_usage.html and/or https://www.sc-best-practices.org/mechanisms/cell_cell_communication.html

R – Follow https://saezlab.github.io/liana/articles/liana_tutorial.html